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Enhanced Elimination of Poisons

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CHAPTER OUTLINE

Poisonings are a major cause of morbidity and mortality, and their treatment is a significant contributor to health care expenditure worldwide. According to the National Poison Data System compiled by the American Association of Poison Control Centers, nearly 2.2 million human poisonings were reported in the United States in 2015.[1] Approximately 78% of reported cases were unintentional (e.g., therapeutic error, bites, stings), 18% were intentional (e.g., deliberate self-poisoning, misuse, abuse), and the remaining were adverse reactions or related to other reasons. Although most reported exposures occurred in children, most fatalities involved adults. Unfortunately, over the past 15 years, human exposures associated with more serious outcomes have increased by 4.3% per year. The classes of drugs accounting for most fatal cases were sedatives, hypnotics, antipsychotics, cardiovascular agents, opioids, stimulants, and other recreational drugs. Table 67.1 shows the 2015 U.S. statistics of the incidence and outcomes related to various poisons that may require the care of a nephrologist.

All acute exposures to potentially toxic xenobiotics should be considered life-threatening until a complete risk assessment has been performed.[2] The general approach to the poisoned patient necessitates prompt resuscitation and stabilization, clinical and laboratory evaluation, antidote administration, and gastrointestinal decontamination, if appropriate, which are outlined elsewhere.[2,3] Although the prognosis of most toxic exposures is excellent with supportive measures, a minority of poisonings can benefit from treatments that actively enhance elimination (Fig. 67.1). Elimination enhancement modalities can be divided between corporeal treatments, which augment physiologic processes, and extracorporeal treatments (ECTRs), which require an artificial device located outside the body.[4]

Evidence-based treatment recommendations from randomized trials are lacking for many poisonings due to infrequent presentations, heterogeneous baseline characteristics, complexity with regard to consent in urgent conditions, and potential ethical issues comparing treatment by ECTRs with a control. The current evidence base is mostly derived from retrospective observational cohorts, human case reports, and animal studies, with each having significant limitations.

Fortunately, consensus-based expert recommendations derived from published data are now becoming available for both corporeal[5,6] and extracorporeal treatments[7–19] and have been endorsed by international toxicology and nephrology societies. To treat poisonings efficiently, attending physicians must know the characteristics of the poisons, their clinical effects, pharmacokinetics, and potential removal by various techniques. The purpose of this chapter is to review the fundamental concepts of poison elimination, available elimination enhancement modalities, and poisons for which the expertise of nephrologists is most likely needed.

OVERVIEW OF CORPOREAL TREATMENTS FOR ENHANCED ELIMINATION OF POISONS

FORCED DIURESIS

Physiologic mechanisms involved in the elimination of xenobiotics by the kidney include the following: (1) glomerular filtration; (2) active secretion in the proximal tubule; and (3) reabsorption in the distal tubule. The intended goal of forced diuresis is to induce a urine output in excess of what is considered physiologic (≥4–5 mL/kg per hour) to enhance the renal elimination of poisons. Historically, forced diuresis is performed through volume expansion with isotonic fluids (0.9% NaCl or lactated Ringer solution), with or without concomitant diuretics. Results of early studies have failed to

Table 67.1 American Association of Poison Control Centers' Statistics for Exposures of Common Dialyzable Toxins

Poison	Total Exposures[a]	Deaths
Acetaminophen	120,156	111
Barbiturates	1997	2
Carbamazepine	3574	0
Ethylene glycol	6178	22
Gabapentin	17,702	2
Isopropanol	16,538	2
Lithium	7306	4
Metformin (biguanides)	8733	8
Methanol	2117	16
Paraquat	113	3
Phenytoin	2556	2
Salicylates	29,504	21
Theophylline	146	2
Valproic acid	7928	4

[a]Includes coingestions and mixed formulations.

Adapted from Mowry JB, Spyker DA, Brooks DE, et al. 2015 Annual Report of the American Association of Poison Control Centers' National Poison Data System (NPDS): 33rd Annual Report. *Clin Toxicol (Phila).* 2016;54(10):924–109.

show any significant benefit of forced diuresis. Unfortunately, forced diuresis is also associated with complications, such as volume overload, pulmonary edema, cerebral edema, and electrolyte disturbances (e.g., hyponatremia, hypokalemia). At present, forced diuresis is never recommended in the management of acute poisonings. However, aggressive volume repletion remains warranted for some poisons to correct hypotension and/or to overcome tubular reabsorption of some offending agents (e.g., lithium) during volume contraction.

URINARY ALKALINIZATION

Urinary alkalinization is based on the concept of what is termed "ion trapping." The goal of manipulating urinary pH is to create an ionized form of the poison, which is less amenable to passive reabsorption, because ionized particles are less lipid-soluble. The ionized poison then becomes "trapped" in the renal tubular lumen and is eliminated in the urine. The dissociation of a weak acid or base into its ionized state is determined by its dissociation constant (pK_a)—that is, the pH at which it is 50% ionized and 50% nonionized. For example, the pK_a of salicylic acid is 3, so when the urinary pH is 3, salicylate exists in a 1:1 ratio of the ionized to nonionized form. Alkalinization of the urine

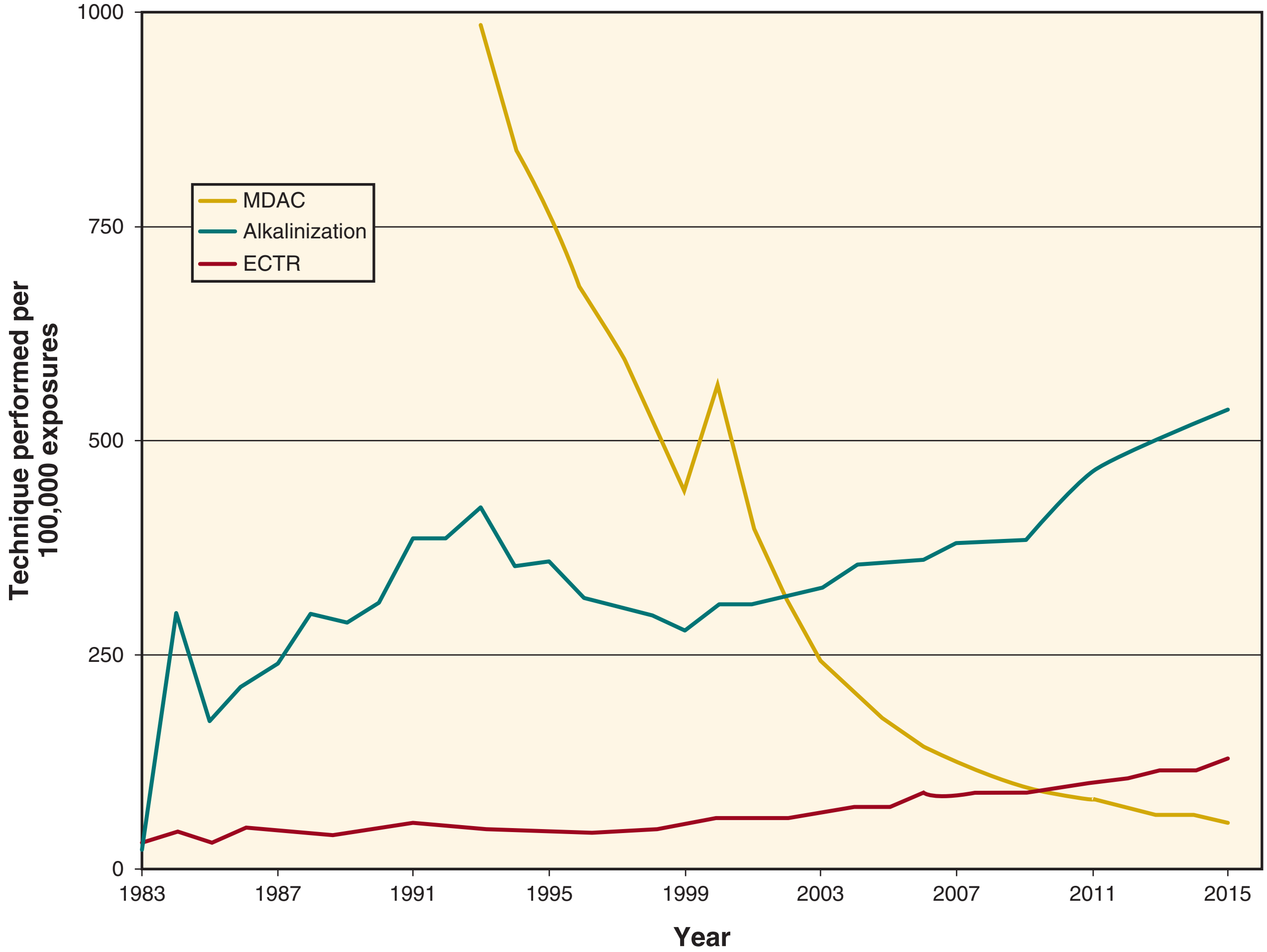

Fig. 67.1 This figure illustrates how often these various elimination enhancement techniques were used in the United States from 1983 to 2015. *ECTR,* Extracorporeal treatment; *MDAC,* multiple-dose activated charcoal.

to a pH of 7.4 increases the ratio to 20,000:1 in favor of the ionized form, which is more readily eliminated in the urine.

The clinical efficacy of urine alkalinization depends on the relative contribution of kidney clearance to the total body clearance of active poison. If only 1% of the ingested poison is excreted unchanged in the urine, even a 10-fold increase in renal elimination will have no clinically significant effect.[6] Criteria that determine whether a poison is amenable to urinary alkalinization include the following: (1) it is eliminated unchanged by the kidneys; (2) it is not strongly protein-bound; (3) it is distributed primarily in the extracellular fluid compartment; and (4) it is weakly acidic (i.e., with a pK_a of 3.0–7.0). Urine alkalinization is generally used to enhance the excretion of salicylates and phenobarbital, but can also be used for chlorpropamide, 2,4-dichlorophenoxyacetic acid, mecoprop (methylchlorophenoxypropionic acid), diflunisal, fluoride, and methotrexate (Table 67.2).[6] For salicylate poisoning, urine alkalinization reduces the elimination half-life from 19 to 5 hours.[20]

Urine alkalinization requires the administration of intravenous sodium bicarbonate by the infusion of a bolus of 1 or 2 ampoules of 50 mL of 8.4% sodium bicarbonate, followed by 2 or 3 ampoules added to each liter of 5% dextrose in water. The rate of infusion should be adapted to the volume status and cardiac condition of the patient but can be as high as 250 mL/h. Contraindications include severe kidney disease, pulmonary edema, and cerebral edema. Serum electrolytes and urinary pH must be closely monitored (every 4 hours). The target urine pH is between 7.5 and 8.5 while maintaining a serum blood pH no higher than 7.55. Complications of urine alkalinization include hypokalemia, hypocalcemia, hypernatremia, fluid overload, and pulmonary and cerebral edema, as well as metabolic alkalosis. The degree of hypokalemia may be profound because of intracellular potassium shifts and urinary potassium losses. Moreover, normokalemia is a prerequisite for urine alkalinization to be effective; in the setting of hypokalemia, potassium is reabsorbed at the collecting tubule in exchange for a hydrogen ion. Therefore, if hypokalemia remains uncorrected during urine alkalinization, not only is the nephron unable to produce alkaline urine, but the patient is also at a higher risk of developing alkalemia. Carbonic anhydrase inhibitors (e.g., acetazolamide) can alkalinize urine but also create metabolic acidosis; this can exacerbate toxicity by increasing the proportion of the poison that is nonionized in the blood, which increases its intracellular distribution and toxicity, and thus these are never recommended.

In the past, urinary acidification, aiming for a urinary pH less than 6.0, was used to enhance renal elimination of weak bases such as amphetamines, phencyclidine, and quinine. It is no longer recommended because of lack of efficacy and potential complications.

Table 67.2 Poisons Whose Elimination May Be Enhanced by Corporeal Techniques

Urine Alkalinization	Multiple Doses of Activated Charcoal	Sodium Polystyrene Sulfonate	Prussian Blue
Chlorophenoxy herbicides (e.g., 2,4-D, MCPA, MCPP)[a]	Carbamazepine	Lithium	Radiocesium
Chlorpropamide	Colchicine	Potassium	Thallium
Diflunisal	Dapsone		
Fluoride	Digoxin		
Methotrexate	Phenobarbital		
Phenobarbital	Phenytoin		
Salicylates	Quinine		
	Salicylates		
	Theophylline		
	Yellow oleander		

[a]*2,4-D,* 2,4-Dichlorophenoxyacetic acid; *MCPA,* 4-chloro-2-methylphenoxyacetic acid; *MCPP,* methylchlorophenoxypropionic acid (Mecoprop).

FECAL ELIMINATION ENHANCEMENT

Activated charcoal, when given in multiple doses, can enhance the elimination of certain poisons. Ideal properties of poisons amenable to multiple-dose activated charcoal (MDAC) include a low intrinsic clearance, small volume of distribution, and prolonged half-life. MDAC promotes the clearance of poisons by interrupting their enterohepatic circulation, by promoting passive diffusion down a concentration gradient from the intestinal capillaries to the intraluminal gut space, a process often referred as "gut dialysis," and/or by limiting the absorption of controlled-release formulations. Multiple dosage regimens exist and usually include 50 g every 4 hours or 25 g every 2 hours until an improvement in clinical status is seen. Contraindications to the administration of MDAC include an altered level of consciousness with an unprotected airway, protracted vomiting unresponsive to antiemetic therapy, and impaired intestinal function, such as obstruction or ileus. Complications such as aspiration pneumonitis, appendicitis, and bowel obstruction due to charcoal bezoar are very rare, and their incidence increases with the number of doses given. Present guidelines recommend MDAC for poisoning due to carbamazepine, dapsone, phenobarbital, quinine, and theophylline.[5] MDAC is also recommended within 2 hours of salicylate poisoning. In addition, MDAC may be of benefit in select poisonings due to colchicine, cardiac glycosides,[21] or phenytoin (see Table 67.2).[22]

Ion exchange resins may also attract poisons from the gut capillaries to the lumen. Sodium polystyrene sulfonate is used to treat hyperkalemia, and there is current evidence that it can also reduce lithium's half-life.[23] Prussian blue can be used to enhance the fecal elimination of radiocesium and thallium (see Table 67.2).[24]

Whole-bowel irrigation (WBI) consists of the administration of electrolyte-balanced polyethylene glycol (macrogol) solution (up to 2 L/h) to induce diarrhea until the rectal effluent is clear to remove poisons from the gastrointestinal tract. WBI may be considered in specific circumstances, such as massive ingestions of sustained-release or enteric-coated drugs (e.g., salicylate) in a cooperative patient, massive ingestions of toxins not adsorbed by activated charcoal (e.g., iron or lithium tablets), or illicit drugs from body packers. Contraindications include intestinal perforation or obstruction, significant gastrointestinal hemorrhage, and protracted vomiting. WBI may interfere with the adsorptive effect of MDAC if administered concomitantly, but whether this compromises overall elimination is uncertain.

PRINCIPLES AND FACTORS INFLUENCING POISON REMOVAL DURING EXTRACORPOREAL TREATMENTS

The elimination of a poison depends on its physicochemical and pharmacokinetic properties, as well as the extracorporeal technique used. Extracorporeal elimination of a poison is only significant if the following conditions are present: (1) it can be extracted from the plasma compartment, (2) extracorporeal clearance contributes to a significant proportion of total clearance; and (3) if the poison exerts its toxicity outside of the plasma compartment, a significant proportion of its body stores can be eliminated during the treatment. The first condition depends on the molecular size, water solubility, and protein binding of the poison because these correlate to its extraction ratio (ER) and extracorporeal clearance (CL_{ECTR}). ER can be calculated as (A – V)/A, where A represents the inflow (or prefilter) plasma concentration and V represents the outflow (or postfilter) plasma concentration. An extraction ratio of 1.0 implies complete removal of a substance from the plasma after a single pass through the extracorporeal circuit (i.e., V = 0). Extracorporeal plasma clearance may be calculated as

$$CL_{ECTR} = Q_B \times (1 - Hct) \times ER$$

where Q_B is the blood flow and Hct is the hematocrit. CL_{ECTR} can also be calculated by quantifying the amount of poison in the spent ultrafiltrate and/or dialysate over a period of time and dividing by its averaged plasma concentration during the same time frame. The second condition relates to its endogenous clearance, and the third depends on its volume of distribution (V_D; see later).

Modality-specific factors to consider include the process of poison removal (e.g., diffusion, adsorption, convection, centrifugation; Table 67.3), as well as the parameters chosen for a specific technique, such as the dialysis membrane (dialyzer surface area, membrane pore structure), blood flow rate, and effluent flow rate.[25] These are summarized later, after "Poison-Related Factors."

POISON-RELATED FACTORS

MOLECULAR SIZE

Poisons with a molecular weight below 1000 Da will be removed by any of the processes noted but are removed best by diffusion and convection. Solutes in excess of 1000 Da will be more easily removed by convection, adsorption, or centrifugation,[26] although modern intermittent hemodialysis (IHD) using diffusion alone can now clear poisons with an upper limit of approximately 10,000 Da (see Table 67.3). Convection from hemodiafiltration (HDF) and continuous renal replacement therapy (CRRT) can remove poisons with molecular weight up to 50,000 Da. Poisons with very high molecular weights (>100,000 Da) can only be removed by adsorption or centrifugation.

PROTEIN BINDING

The degree of protein binding will also determine its removal. Hemofiltration and hemodialysis can remove only unbound poison because the poison–protein complex size exceeds the pore size of the hemofilter or dialyzer. Diffusion (IHD) and convection (HDF, CRRT) can remove poisons with protein binding up to 80%, with a few exceptions. Hemoperfusion, however, may be more effective in poisons with protein binding up to 90% to 95%, because binding to the adsorbent (activated carbon or, less commonly, a resin) competes with

Table 67.3 Summary of Extracorporeal Treatments[a]

Treatment	Process	Molecular Weight Cutoff (Da)	Protein Binding Cutoff	Relative Cost	Complications	Comments
Albumin dialysis	Diffusion, adsorption	<60,000–100,000	<95%	++++	++	Liver replacement support
CRRT	Convection and/or diffusion	<10,000–50,000	<80%	++	+	Correction of uremia and acid-base and E+ disorders
Exchange transfusion	Centrifugation, separation, filtration	None	None	++	++	Easier than other ECTRs in neonates; correction of hemolysis
Hemodialysis	Diffusion	<10,000	<80%	+	+	Correction of uremia and acid-base and E+ disorders
Hemofiltration	Convection	<50,000	<80%	++	+	Correction of uremia and acid-base and E+ disorders
Hemoperfusion	Adsorption	<50,000	<95%	++	+++	Saturation of cartridge requires changes
Peritoneal dialysis	Diffusion	<500–5000	<80%	++	++	Low efficacy
Plasma exchange	Centrifugation, separation, filtration	<1,000,000	None	+++	+++	

[a]All extracorporeal treatments above are less likely to be useful for poisons that have a high V_D or a high endogenous clearance.
CRRT, Continuous renal replacement therapy; *E+,* electrolyte; *ECTR,* extracorporeal treatment; +, low; ++, moderate; +++, high; ++++, very high.

binding to plasma proteins. The clearance therefore depends on the affinity of the poison for the adsorbent.

The degree of protein binding can be influenced by acute alterations in poison or protein concentration and the presence of different pathologic states.[27] For example, in the context of hypoalbuminemia, there is less protein available to bind poison. As such, the concentration that is free (unbound) is higher with hypoalbuminemia, which will result in increased removal by ECTRs. Similarly, accumulation of organic acids in uremia leads to a reduction in binding sites for some xenobiotics (e.g., salicylates, warfarin, phenytoin), which also increases the unbound concentration and favors removal by ECTR. Furthermore, in toxic concentrations, there may be saturation of the protein binding sites (e.g., valproic acid, salicylate, 4-chloro-2-methylphenoxyacetic acid [MCPA]), increasing the fraction of poison that is unbound in relation to its total concentration, which also increases the amount that is amenable to removal by ECTR.[28]

ENDOGENOUS CLEARANCE

Extracorporeal removal will not be useful if the endogenous clearance of a poison via its metabolism and native elimination routes far outweighs the clearance provided by an ECTR.[29] The endogenous clearance should usually be less than 4 mL/min per kg (or <200 mL/min) for extracorporeal removal to be considered potentially beneficial. This explains why hemodialysis is not indicated for poisons such as cocaine or toluene, which have very high endogenous clearances. Additionally, significant impairment in endogenous metabolic and/or elimination pathways may increase the relative efficacy of extracorporeal treatments; for example, the renal clearance of metformin decreases from 600 to 0 mL/min in anuric acute kidney injury (AKI).

VOLUME OF DISTRIBUTION

The V_D of a drug is the apparent volume into which a poison is distributed at equilibrium before clearance occurs. The V_D is calculated by dividing the total drug in the body by its concentration. This mathematical relationship assumes that the body is a single homogenous compartment of water into which the drug is distributed.

Drugs that distribute extensively in tissues (e.g., tricyclic antidepressants) have a high V_D; conversely, drugs that distribute in total body water, such as methanol, have a low V_D (<1 L/kg). Poisons exclusively confined to the blood compartment (e.g., mannitol) have a V_D of 0.06 L/kg. Because ECTRs can only remove poisons in the intravascular space, they will not be useful to eliminate poisons with a high V_D. For example, due to its low protein binding (25%) and relatively low molecular weight (780 Da), digoxin easily crosses the dialyzer; however, because of its high V_D (7 L/kg), less than 5% of the total body burden of digoxin will be removed in a 6-hour dialysis session. Many publications still erroneously conclude that a poison with a high V_D is amenable to extracorporeal clearance, solely based on a high clearance or rapid reduction of serum concentrations.[30–32]

ECTR may be considered for very toxic xenobiotics with a high V_D (e.g., thallium) only if a patient presents early after exposure. In this case, a high proportion of the poison has not yet distributed into various tissues or there may be ongoing absorption and so may successfully be removed by an ECTR.[33]

AVAILABLE EXTRACORPOREAL TREATMENTS TO ENHANCE ELIMINATION OF POISONS

Numerous techniques can facilitate poison removal. These can be classified by their mechanism of removal: diffusion (hemodialysis, peritoneal dialysis), convection (hemofiltration), adsorption (hemoperfusion), and centrifugation (therapeutic plasma exchange).[34,35] The features of these treatments are summarized in Table 67.3.

INTERMITTENT HEMODIALYSIS

During IHD, the poison diffuses from the blood compartment to a dialysate flowing in a countercurrent direction, separated by a semipermeable dialysis membrane. The principles that dictate solute removal in IHD also apply to poison elimination (see Chapter 63 for a detailed discussion). The characteristics of a xenobiotic that favor efficient poison removal by IHD are a low molecular weight (<5000–10,000 Da), low V_D (<1–2 L/kg), low protein binding (<80%), and low endogenous clearance (<4 mL/min per kg).

Specific components of the dialysis system will influence poison clearance. These components include the type of dialysis membrane, its surface area, and the blood and dialysate flow rates (Box 67.1). Poison elimination is limited by the pore size of the dialysis membrane. However, even if the poison size is smaller than the cutoff of the membrane, larger molecules do not diffuse as freely as smaller ones, so clearance is reduced with larger molecules. New synthetic high-flux membranes and catheters now allow the removal of larger poisons considered nondialyzable 20 years ago.[26] Increasing the dialysate flow rate and, more importantly, the blood flow rate, will result in greater diffusion and elimination of the poison.[35] In poisoning, a filter with a large surface and maximal blood and dialysate flow rates should be used unless a contraindication is present (e.g., concern for disequilibrium syndrome).[35]

IHD has several distinct advantages over other extracorporeal modalities in the management of acute poisonings because it rapidly removes poisons and corrects metabolic disequilibria associated with severe poisonings, such as AKI, volume overload, acid–base abnormalities, electrolyte disturbances, and even hypothermia.[36] IHD is also the most available ECTR, the least expensive, and the quickest ECTR to implement.[37]

Box 67.1 Factors That May Enhance Poison Clearance During Hemodialysis

- Larger surface area of dialysis membrane
- High-flux dialyzer
- High blood and dialysate flows
- Increased ultrafiltration rate (with replacement solution)
- Increased time on dialysis
- Reduced recirculation
- Two dialyzers in series
- Two distinct extracorporeal circuits

The most common acute complication of IHD is systemic hypotension, and this is most relevant for patients with AKI or end-stage renal disease (ESRD) who require fluid removal. Because poisoned patients rarely require net ultrafiltration, the true incidence of dialytic hypotension in the toxicology setting is unknown but is more likely related to the effect of the poison, rather than to the dialysis treatment itself. Although hypotension itself is often a marker of toxin severity and can be an indication for ECTR, it is acknowledged that hypotension may limit the ECTR's performance.

HEMOPERFUSION

During hemoperfusion (HP), blood circulates through an extracorporeal circuit equipped with a charcoal or resin cartridge onto which the poison can be adsorbed.[38] Compared with diffusion, adsorption is not as limited regarding the molecular size, lipid solubility, or protein binding of the poison. Alcohols and most metals, however, are poorly adsorbed to HP columns. Although certain exchange resins (e.g., XAD-4) are effective in the removal of organic solutes and nonpolar, lipid-soluble drugs, they are no longer available in the United States.

Current HP devices have improved their biocompatibility over the years; the coating of the sorbent material minimizes direct contact between the adsorbent and blood constituents, thereby greatly reducing the risk of embolization without impairing its adsorptive capacity. The circuit requires more generous systemic anticoagulation than dialysis. Blood flows are limited to 350 mL/min to minimize the risk of hemolysis.[39]

HP-associated complications are mostly related to the nonselective adsorption of certain cells and molecules, such as platelets (≅30%–50%), white blood cells (≅10%), serum fibrinogen, fibronectin, calcium, and glucose.[40,41] From a practical point of view, thrombocytopenia, hypoglycemia, and hypocalcemia can occur within hours after treatment initiation. Although these complications were more common with older devices and are reversible and treatable, HP still carries a higher risk of complications than IHD.[42] Furthermore, a cartridge costs several-fold more than a dialyzer and needs to be replaced every 3 to 4 hours due to saturation.

Twenty years ago, clearances of many poisons were higher with HP compared with IHD. However, with recent technologic advancements in IHD, these results have improved. For example, clearances of theophylline and phenobarbital by HP or IHD are now considered comparable.[38,42,43] In addition, HP does not correct electrolyte and acid–base disturbances, cannot remove fluid, and is less readily available. For these reasons, IHD is generally preferred in most settings where HP is indicated.[38] These considerations are reflected by recent trends in the choice of ECTR for poisonings (Fig. 67.2).[44–46]

HEMOFILTRATION

During hemofiltration (HF), solute and solvent are removed by solvent drag, a process known as "convection," and replaced by a physiologic solution. Ultrafiltration is dependent on the sieving coefficient of the membrane. The sieving coefficient is the ratio of a solute concentration in the ultrafiltrate to its respective concentration in the plasma. A solute that freely crosses the membrane has a sieving coefficient of 1, whereas a sieving coefficient of 0 means that the solute cannot cross the membrane. Drug elimination by HF depends on factors similar to those regulating diffusion. However, convective

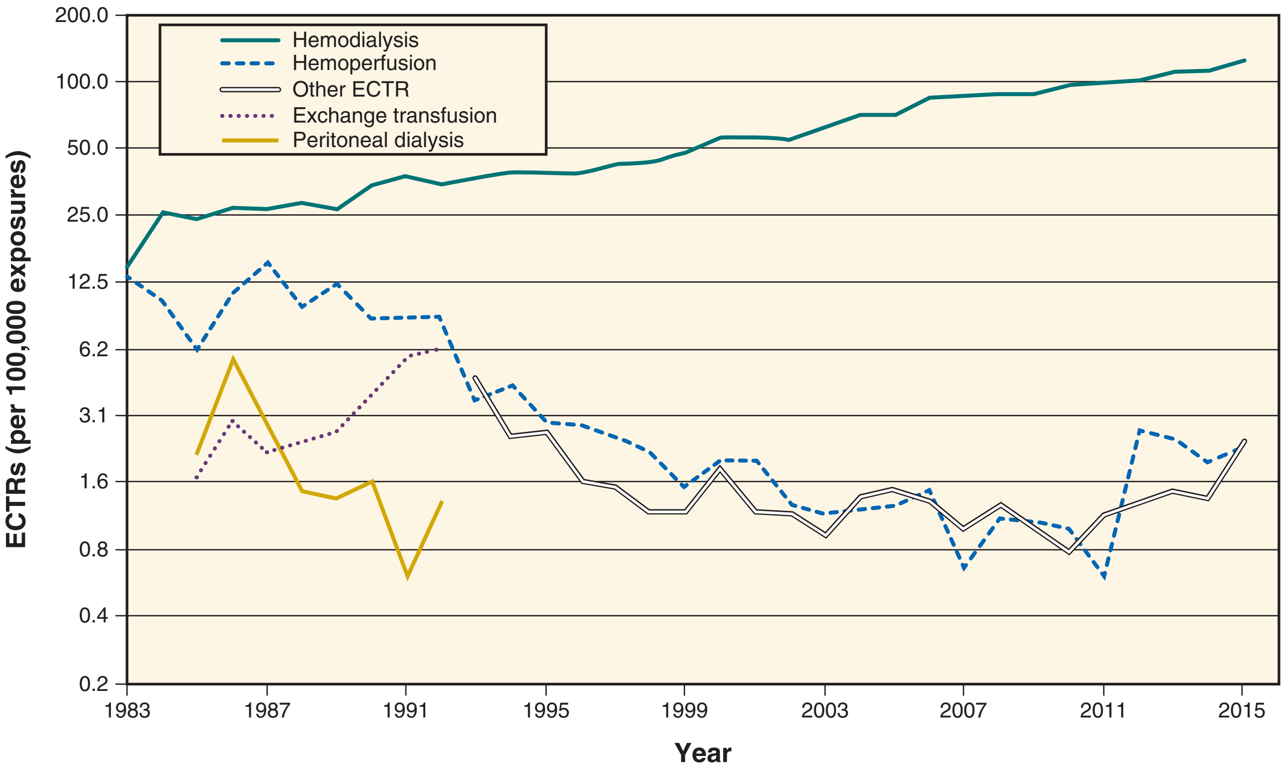

Fig. 67.2 This figure illustrates how often extracorporeal treatments were used in the United States from 1983 to 2015. *ECTR,* Extracorporeal treatment.

transport also allows the removal of larger toxins—that is, having a molecular weight of up to 50,000 Da.[35] Because most poisons have a low molecular weight (<1000 Da), HF does not offer an advantage over IHD in the removal of most poisons.

COMBINED THERAPIES

Clinicians sometimes combine more than one mechanism to optimize poison removal. For example, adsorption (HP) and diffusion (IHD) are sometimes used concurrently, in series, to maximize poison clearance. Although there is evidence that this combination approach may yield better results, the amount removed by the two procedures in series is usually not additive of the effect when either is used solely and may also increase the incidence of complications. Diffusion and convection (intermittent HDF) are also commonly combined.

CONTINUOUS RENAL REPLACEMENT THERAPY

Most ECTRs can be offered intermittently or continuously. CRRTs are particularly popular in the critical care setting to manage AKI, and they are available in various forms (see Chapter 65): continuous venovenous hemodialysis (CVVHD), continuous venovenous hemofiltration (CVVH), and continuous venovenous hemodiafiltration (CVVHDF). Their role in the management of acute poisonings remains uncertain.[47] The solute clearance rate with continuous modalities is usually much lower than with intermittent modalities because of lower blood, dialysate, and ultrafiltrate flow rates. In AKI, continuous modalities are often chosen for fluid removal because of their better tolerability in hemodynamically unstable patients. However, in poisoned patients, because net ultrafiltration is rarely required, and because the removal of the offending agent is urgent, intermittent therapies are usually the modality of choice. CRRT may be used after IHD to avoid a sudden reincrease in poison concentration (so-called rebound); however, the benefit of this practice remains to be determined (see later).

Sustained low-efficiency dialysis (SLED) is a hybrid technique between IHD and CRRT.[48,49] Although used in anecdotal reports,[50] it provides a lower poison clearance than IHD over a same period of time.

PERITONEAL DIALYSIS

Peritoneal dialysis (PD) clears substances by diffusion but has a limited role in acute poisoning because the maximum clearance obtainable is usually 10 to 15 mL/min (<10% of the clearance achievable by IHD).[34] PD has no advantages over other extracorporeal treatments in poisoning.

THERAPEUTIC PLASMA EXCHANGE AND PLASMAPHERESIS

Plasmapheresis involves the withdrawal of venous blood and separation of plasma from blood cells by filtration or centrifugation and reinfusion of cells plus autologous plasma or another replacement solution (see Chapter 66). In therapeutic plasma exchange (TPE), the removed plasma is discarded and replaced by 5% albumin or fresh-frozen plasma. During TPE, clearance is limited by the plasma removal rate, which cannot exceed 50 mL/min.[34,51] The role of TPE in the treatment of acute poisoning is not well defined, but this method should only be considered for highly protein-bound (>95%) or very large poisons over the accepted cutoffs for HP or HF (>50,000 Da), if available. Drugs for which TPE may increase clearance in poisoning include cisplatin,[52] L-thyroxine,[53] and vincristine.[54] TPE has also been reported to treat near-fatal infusion reactions to rituximab.[55] Complications of TPE include those associated with installation of the vascular access, bleeding, hypocalcemia, and hypersensitivity reactions to the replacement solution.[56,57]

EXCHANGE TRANSFUSION

During exchange transfusion (ET), there is removal of whole blood or red blood cells (by apheresis); this is used to remove the patient's red blood cells and replace them with transfused blood products. Its role in poisoning is unclear but may be considered in poisons causing massive hemolysis (e.g., sodium chlorate or nitrite poisoning in a patient with glucose-6-phosphate dehydrogenase [G6PD] deficiency) or in infants because it is technically less cumbersome to use than IHD in this population.

EXTRACORPOREAL LIVER ASSIST DEVICES (ALBUMIN DIALYSIS)

Albumin dialysis is a relatively new extracorporeal treatment that can be used to replace liver function in fulminant hepatitis or severe cirrhosis, often as a bridge to transplantation. There are three different types of albumin dialysis. Single-pass albumin dialysis (SPAD) is a technique similar to IHD or CRRT but with albumin added in the dialysate. The albumin is discarded after its contact with the filter. The Molecular Adsorbents Recirculation System (MARS) is identical to SPAD, but the albumin-enhanced dialysate (with the adsorbed xenobiotics) is recycled after going through a dialysis filter, resin, and charcoal cartridge. The Prometheus system combines albumin adsorption with high-flux IHD after selective filtration of the albumin fraction through a polysulfone filter.

Theoretically, these devices can remove albumin-bound xenobiotics and endogenous substances (e.g., bile acids, bilirubin) better than the classic diffusive and convective techniques. However, preliminary data have not shown their superiority in terms of clearance of theophylline, valproic acid, or phenytoin.[58–60] An application of extracorporeal liver assist devices in toxicology may include toxin-induced hepatotoxicity, especially from *Amanita* mushroom or acetaminophen.[61–64] The precise role of liver assist devices in poisoning is currently unclear due to their relatively limited clearance, high cost, complication rates, and limited availability. Table 67.3 summarizes the various extracorporeal treatments available for poison removal.

GENERAL INDICATIONS FOR EXTRACORPOREAL REMOVAL OF POISONS

The EXTRIP (EXtracorporeal Treatment In Poisoning; http://www.extrip-workgroup.org/) work group[65,66] has completed guidelines for the use of blood purification for several key toxins.[7–19] These recommendations help standardize management for these types of poisonings and propose directions for future research.[66] The decision to initiate any form of blood purification must take into account the clinical status, benefit expected from extracorporeal treatments, and poison-related factors.

Absolute indications for ECTR include the following (all must be present)[36]:

1. The poison exposure will result in severe toxicity. The exposure to a specific poison must be significant enough to warrant the cost and complications associated with ECTR. Obviously, a patient with life-threatening clinical signs (e.g., repeated seizures, respiratory depression, dysrhythmias) will be categorized as severe. A proper risk assessment of the exposure, which includes close collaboration with a poison control center, may help estimate the risk for any specific patient. In some cases, a poison exposure produces delayed effects (e.g., methanol, paraquat, acetaminophen), and monitoring of their concentration might predict future clinical compromise and would prompt prophylactic ECTR (i.e., before the appearance of toxic symptoms).
2. There is an absence of preferable alternative treatments. Antidotes may amend or prevent the apparition of toxicity related to a poison, so ECTR then becomes less crucial or indicated. This is the case for acetaminophen poisoning, when *N*-acetylcysteine is available. Because ECTR is somewhat invasive and may require transfer to a specialized center, its cost and benefit should be weighed against those of the antidote.
3. The ECTR must be able to remove the poison. (see earlier). To decide whether blood purification is indicated for a specific poisoning, the clinician must anticipate which benefits are expected from the procedure; some exposures can cause death (e.g., salicylates at high ingestions, paraquat), whereas others may cause irreversible tissue damage (e.g., blindness from methanol). The advantages of ECTR in those circumstances would largely outweigh the costs and complications of the procedure. In other situations, the poisoning itself may not cause irreversible injury, but the patient may be subjected to prolonged coma and immobilization, requiring mechanical ventilation and close surveillance in the intensive care unit (ICU), as seen in poisonings causing central nervous system (CNS) depression (e.g., barbiturates, anticonvulsants).
4. Finally, there may be situations in which ECTR will likely not affect outcomes but may reduce ICU or hospital length of stay and associated costs (e.g., dialysis versus fomepizole alone in a methanol-poisoned patient without metabolic acidosis).[67] The clinician should therefore assess the risks of the specific exposure and assess the cost-effectiveness of an ECTR in this context. Complications associated with ECTR are minimal and are usually limited to the insertion of a vascular access (which can be minimized with ultrasonographic guidance).[68] ECTR can also potentiate the elimination of certain antidotes (e.g., ethanol, *N*-acetylcysteine)[69] and precipitate withdrawal symptoms if drug concentrations fall below the therapeutic range (e.g., phenobarbital in a non-naïve patient).[70] Costs of a single dialysis, including equipment, nursing, and physician fees, are minor compared with the cost of a day in the ICU.[37] In the absence of any clinical outcome data, studies should demonstrate, at a minimum, significant drug removal.

TECHNICAL CONSIDERATIONS

Patients presenting with poisonings are different from patients with AKI or ESRD. Therefore, the prescription of any extracorporeal treatment for the purpose of poison removal should reflect these differences.

VASCULAR ACCESS

A double-lumen central catheter is required for administering most forms of ECTR. Because time is usually a concern, a temporary instead of a permanent catheter is preferred, using ultrasound guidance to reduce complications and ensure patency.[71] The femoral line is simpler for insertion and does not require X-ray placement confirmation but increases recirculation compared with subclavian or right jugular insertion sites.[72] However, there is less catheter malfunction with a femoral catheter than with a left jugular catheter.[73] There has been some experience on the use of dual-catheter sites and circuits to maximize poison clearance.[74,75]

CHOICE OF HEMODIALYZER, FILTER, AND ADSORBER

For IHD, high-flux, high-efficiency dialyzers with the largest surface area should be used. The dialyzer or hemofilter should have a molecular size cutoff above that of the poison to be removed. With respect to HP, the only column available in the United States is the Gambro Adsorba 300C (Baxter Healthcare, Wayne, PA), a coated, activated, charcoal cartridge.[38]

ANTICOAGULATION

Heparinization of the dialysis circuit is usually done with unfractionated heparin or low-molecular-weight heparin to prevent clotting and maintain patency of the circuit. In patients at high risk of bleeding, heparin can be substituted by saline flushes. For HP, heparin is also used to reduce the risk of hemolysis[39] and is usually required in larger quantities than for IHD.

BLOOD, DIALYSATE, AND EFFLUENT FLOW

These should be maximized to increase clearance according to the capacities of the machine.[35]

DIALYSATE COMPOSITION

As mentioned, poisoned patients may not present with the same metabolic disturbances as those with AKI or ESRD. The sodium, bicarbonate, potassium, calcium, and magnesium concentrations in the dialysate (or replacement fluid) need to be adjusted according to their serum concentrations. Phosphate may also be added to the dialysate or replacement fluid to avoid hypophosphatemia. It is also recommended that periodic measurements of serum biochemistry be performed and the content of the dialysate modified accordingly.

DURATION OF EXTRACORPOREAL TREATMENT

A single 6-hour, high-efficiency ECTR will usually suffice for most poisonings for which ECTR is indicated. When significant toxicity is present or suspected to be prolonged, there is little risk to extend a treatment for several additional hours. However, HP cartridges usually need to be replaced after 3 to 4 hours because of saturation. Obviously, a longer period of treatment will be required if a less efficient therapy (e.g., CRRT, peritoneal dialysis, SLED) is used.

PATIENT DISPOSITION

Many poisoned patients die prior to the initiation of dialysis.[76] If the risk assessment for a suspected toxic exposure suggests that a patient may require dialysis, prompt communication with a dialysis unit and preemptive transfer may be required, even if the patient does not yet meet the criteria for blood purification. Because a significant delay may occur between the time a decision is taken to perform ECTR and the time when it is initiated,[37] a dialysis nurse should be rapidly contacted and a temporary dialysis catheter installed as early as possible. Logistics and clinical status may require transfer of the patient to the ICU.[36]

REBOUND

Rebound is defined as a sudden increase in poison concentration after the discontinuation of ECTR and usually occurs after the cessation of intermittent therapies. Rebound may either be due to the redistribution of poison from deep compartments (e.g., tissues and intracellular space) to the plasma (such as seen with lithium,[77] dabigatran,[78] and methotrexate[79]), especially in poisons with a large V_D, or due to ongoing absorption of the poison. In the former case, elevation of the serum concentration might reflect a concomitant decrease of the poison concentration from the toxic compartments (e.g., lithium) and so may even be beneficial.[80] In the latter situation, the increase in serum concentration can cause a recurrence of toxic symptoms. If rebound is thought to be worrisome, a clinician may choose to repeat an intermittent session, switch to a continuous therapy, or extend the intermittent therapy longer than the typical 4- to 6-hour treatment duration.[81] Following ECTR, serial poison concentrations and clinical status should be monitored for a period long enough to account for redistribution (≅12–24 hours) or ongoing absorption. The catheter should remain in place until the physician is convinced that additional sessions are unnecessary.

POISONS AMENABLE TO EXTRACORPOREAL ELIMINATION

In the large majority of poisoning cases, ECTRs are not required. Drugs or poisons that are most commonly responsible for poisoning-related fatalities (e.g., tricyclic antidepressants, short-acting barbiturates, stimulants, "street drugs") are not effectively amenable to extracorporeal removal,[82] or severe toxicity and death occur rapidly, which limit the opportunity to initiate ECTR. The remainder of this chapter focuses on the clinical characteristics of key poisons for which ECTR may be considered. Physicochemical characteristics of major xenobiotics, including toxic alcohols, are shown in Table 67.4; the most common indications for ECTR in the context of poisoning are shown in Fig. 67.3 (U.S. data).

TOXIC ALCOHOLS: ETHYLENE GLYCOL, METHANOL, ISOPROPANOL

Toxic alcohols share many similar molecular characteristics and toxicity and will therefore be considered together. They are all colorless. In its pure form, ethylene glycol (EG) is sweet-tasting, which makes it attractive to young children. It is commonly found in antifreeze, radiator fluid, solvents, hydraulic brake fluid, de-icing solutions, detergents, lacquers, and polishes. Methanol, also known as "wood alcohol," has an odor similar to ethanol. It is used as a solvent, as an intermediate of chemical synthesis during various manufacturing processes, or as an octane booster in gasoline. Products that contain methanol include windshield or glass-cleaning solutions, enamels, printing solutions, stains, dyes, varnishes, thinners, fuels, and antifreeze additives for gasoline. Isopropanol (isopropyl alcohol [2-propanolol]) has a bitter taste. It is commonly found in rubbing alcohol, skin lotion, hair tonics, aftershave lotion, denatured alcohol, solvents, cements, and cleaning products. Although the parent alcohols themselves cause minor toxicity (usually no more than moderate inebriation), their metabolites can induce life-threatening toxicity.

Table 67.4 Physicochemical Characteristics and Toxicokinetics of Various Poisons

Poison	Molecular Weight (Da)	Protein Binding (%)	Volume of Distribution (L/Kg)	Endogenous Clearance in Healthy Adults (mL/min per kg)
Acetaminophen	151	20	1	5
Carbamazepine	236	75	1.2	1.3
Ethylene glycol	62	0	0.6	1.8
Isopropanol	60	0	0.6	1.2
Lithium	7	0	0.8	0.4
Metformin	166	5	5	10
Methanol	32	0	0.6	0.7
Methotrexate	454	50	0.8	1.5
Paraquat	186	5	1.0	8
Phenobarbital	232	40	0.7	0.2
Phenytoin	252	90	0.6	0.4
Salicylic acid	138	80[a]	0.2	1.5
Theophylline	180	60	0.5	0.7
Valproic acid	144	90[a]	0.2	0.1

[a]Protein binding saturation occurs at high concentration.

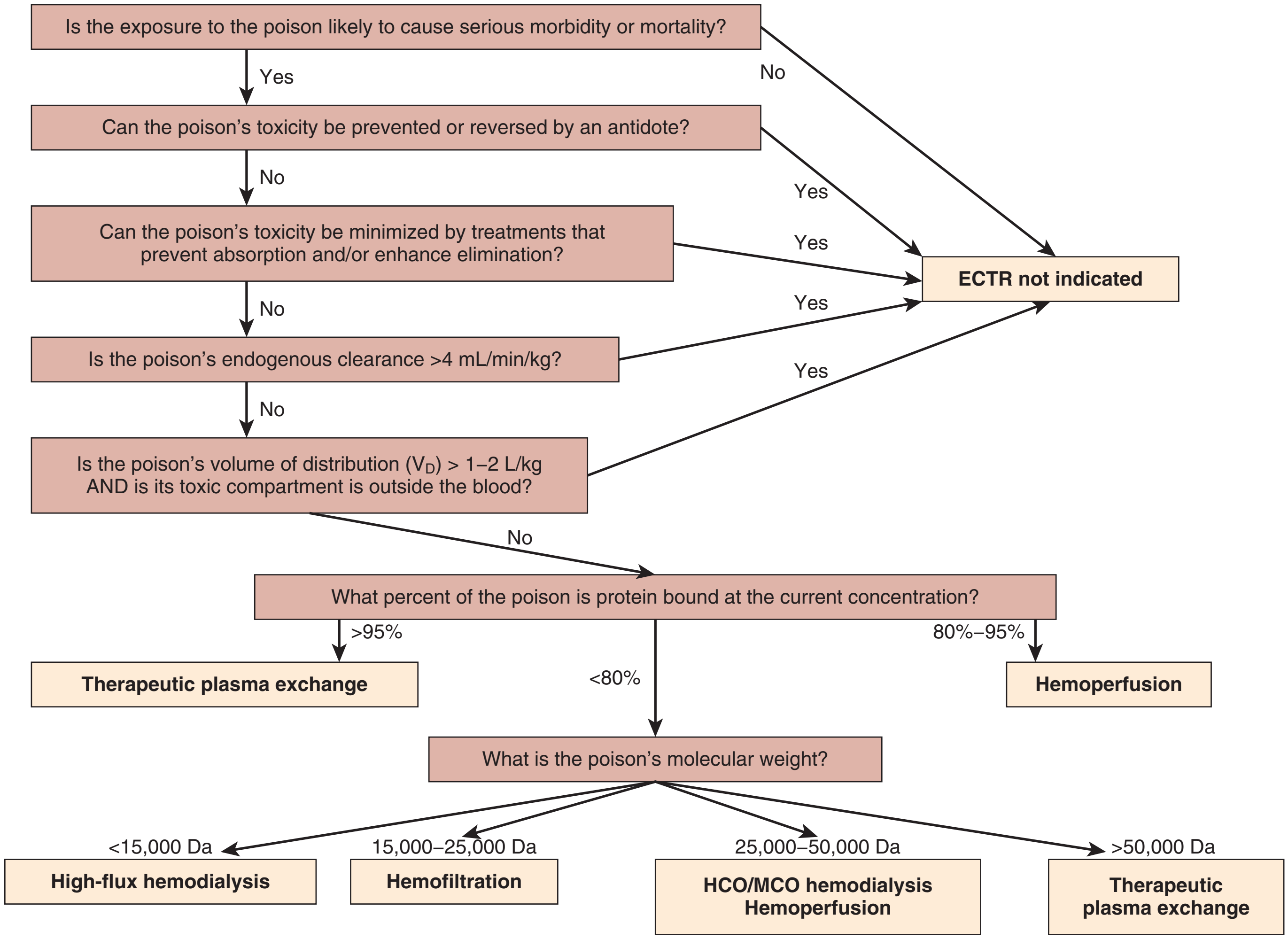

Fig. 67.3 Most common toxin indications for extracorporeal treatments. *ECTR,* Extracorporeal treatment; *HCO,* high cut-off; *MCO,* medium cut-off; V_D, volume of distribution. (Adapted with permission from *Seminars* of dialysis.)

TOXICOLOGY AND TOXICOKINETICS

EG, methanol, and isopropanol are all small unbound molecules that are distributed in total body water ($V_D = 0.6$ L/kg). Intoxication from these alcohols occurs rapidly after exposure and usually results from oral ingestion, although inhalation of vapors[83,84] and cutaneous absorption have been reported, especially in children.[85,86]

Approximately 30% of EG is excreted unmetabolized by the kidneys. The remaining 70% is oxidized to glycoaldehyde by alcohol dehydrogenase (ALDH) in the liver and then rapidly converted to glycolic acid by aldehyde dehydrogenase; this is followed by the slow conversion of glycolic acid to glyoxylic acid, the rate-limiting step (Fig. 67.4A). The end products include oxalic acid, glycine, and oxalomalic acid. Metabolic acidosis results from the formation and accumulation of glycolic acid and glyoxylic acid.[87,88] Lactic acidosis may also be present because the decrease in the NAD^+/NADH ratio promotes the reduction of pyruvate to lactate. However, glycolate also cross-reacts with the lactate assay on some point-of-care blood analyzers, which falsely elevates lactate concentrations. Another mechanism of toxicity of EG is the systemic deposition of precipitated calcium oxalate in various tissues, including the kidneys. Any pure EG ingestion > 0.1 mL/kg requires medical treatment. The potentially lethal dose of EG is approximately 1.4 mL/kg, but deaths have been reported with ingestions as low as 30 mL.

Methanol is metabolized principally by ALDH (85%) to formaldehyde and then rapidly oxidized by formaldehyde dehydrogenase to formic acid (see Fig. 67.4B). Small amounts are excreted unchanged by the lungs (10%) and the kidneys (5%). In a folate-dependent step, formic acid is then transformed to water and carbon dioxide.

Formic acid is the main metabolite responsible for methanol-related toxic symptoms because it induces cellular toxicity due to the inhibition of cytochrome *c* oxidase in cell's mitochondria, interfering with cellular oxidative metabolism.[89] The minimum lethal dose of methanol is estimated at 10 mL, although this is highly variable.[90]

Approximately 80% of isopropanol is metabolized to acetone via ALDH, and the remainder is eliminated unchanged in urine, with very small amounts excreted by the lungs (see Fig. 67.4).[91] Isopropanol displays first-order elimination kinetics, with an elimination half-life of 3 to 8 hours[92,93]; the half-life of acetone is 10 hours.[94,95] ALDH inhibitors markedly increase the half-life of isopropanol but have little effect on that of acetone.[96] Most CNS depressant effects are attributed to isopropanol, whereas acetone likely has a minor effect.[92] The potentially lethal dose of pure isopropanol is reported to be 100 to 250 mL.[97]

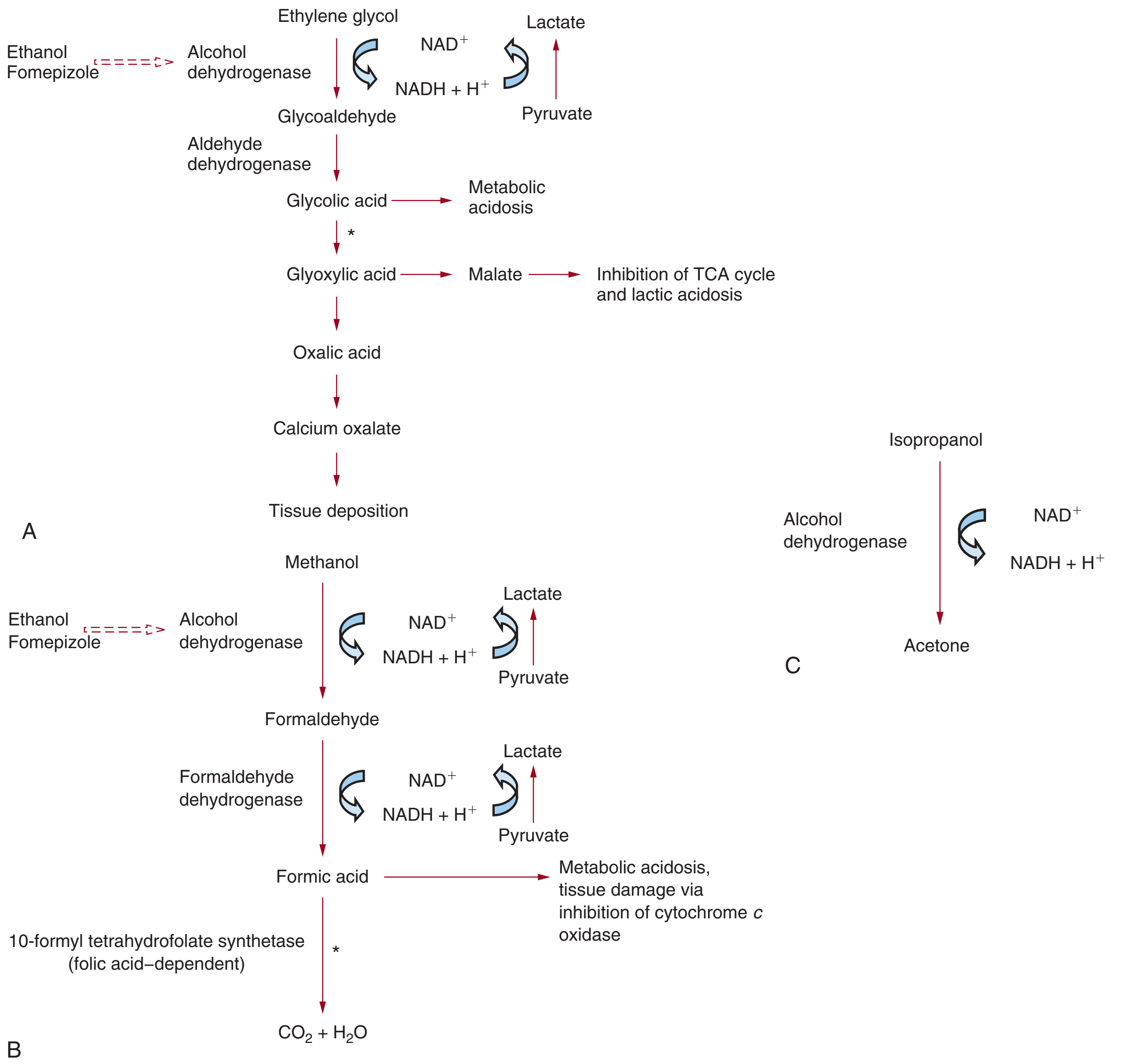

Fig. 67.4 Metabolism of toxic alcohols. (A) Ethylene glycol: The *broken arrow* points to inhibitors of alcohol dehydrogenase; the *asterisk* denotes the rate-limiting step. In the presence of the electron acceptor, nicotinamide adenine dinucleotide *(NAD^+)*, ethylene glycol is oxidized to glycoaldehyde by alcohol dehydrogenase. Aldehyde dehydrogenase then rapidly converts glycoaldehyde to glycolic acid, which is followed by the slow conversion of glycolic acid to glyoxylic acid (the rate-limiting step). The final end products include oxalic acid, glycine, and oxalomalic acid, which are all effectively removed by IHD. (B) Methanol: The *broken arrow* points to inhibitors of alcohol dehydrogenase; the *asterisk* denotes the rate-limiting step. (C) Isopropanol: *IHD,* Intermittent hemodialysis; *4-MP,* fomepizole; *NAD^+,* nicotinamide adenine dinucleotide; *$NADH + H^+$,* reduced form of nicotinamide adenine dinucleotide; *TCA,* tricarboxylic acid.

CLINICAL PRESENTATION

Following ingestion, the severity of the symptoms associated with EG poisoning depends on the ingested dose, concurrent ingestion of ethanol, and timing of treatment.[98] The first symptoms are neurologic and occur 0.5 to 12 hours after ingestion due to the parent alcohol causing inebriation, similar to ethanol but without the typical breath smell. A high osmol gap is present. Altered consciousness may progress to coma and seizures. Cerebral edema, nystagmus, ataxia, myoclonic jerks, and hyporeflexia have been described. New onset of cranial nerve (CN) defects (in particular CN VII) should make a clinician suspect EG ingestion. Gastrointestinal irritation may lead to vomiting, hematemesis, and aspiration pneumonia. Cardiopulmonary symptoms may occur 12 to 24 hours after ingestion; these result from the accumulation of newly formed organic acids. Calcium oxalate crystals deposit in the vasculature, myocardium, and lungs.[99] Hypertension or hypotension, dysrhythmias, myocarditis, pneumonitis, and noncardiogenic pulmonary edema have all been reported.

A severe high anion gap (AG) metabolic acidosis is present, and most deaths occur at this stage. Finally, 24 to 72 hours after ingestion, flank pain, hematuria, crystalluria, and AKI can occur due to the precipitation of calcium oxalate crystals in the kidneys. The pathogenesis of AKI is unclear but may relate to interstitial nephritis, cortical necrosis, direct renal cytotoxicity, or tubular obstruction. However, because the degree of renal injury does not correlate with the extent of calcium oxalate deposition in the kidney, it has been suggested that glycolic acid or other metabolites may be primarily responsible for the development of AKI.[100] Additionally, a low serum calcium concentration may occur and cause brisk deep tendon reflexes and QT interval prolongation.

Methanol intoxication should always be suspected in a patient presenting with neurologic, visual, and gastrointestinal symptoms in the presence of high AG metabolic acidosis with an increased osmol gap. The CNS presentation may include inebriation, headaches, dizziness, nausea, and seizures and may progress to cerebral edema. Methanol can produce a Parkinson-like syndrome by damage to the putamen and subcortical white matter of the basal ganglia.[101,102] Gastrointestinal symptoms include anorexia, nausea, vomiting, gastritis, and abdominal pain from acute pancreatitis. Metabolic acidosis, caused by the accumulation of formate and lactate, can be severe. Vision deficits are the hallmark of methanol poisoning and usually occur 6 to 30 hours after exposure, depending on how much ethanol has been coingested. Vision deficits include blurred vision (flashes or snowstorm), central scotoma, impaired papillary response to light, decreased visual acuity, photophobia, visual field defects, and progression to complete blindness.[103,104] The mechanisms of vision deficits are poorly defined but may relate to the inhibition of mitochondrial function in the optic nerve. Because the optic nerve cells possess few mitochondria and low cytochrome oxidase levels, they are extremely susceptible to the toxic effects of formic acid. Vision deficits are permanent in 25% of cases. Death is usually caused by cardiovascular shock and respiratory arrest.

The diagnosis of isopropanol overdose should be suspected in any patient with altered sensorium, a "fruity" acetone breath, an increased osmol gap without an increase in the AG, and the presence of acetonemia or acetonuria in the absence of hyperglycemia, glycosuria, or acidosis. Isopropanol poisoning mainly affects the CNS, with symptoms ranging from mild inebriation to lethargy, stupor, respiratory depression, and even coma. Isopropanol is also a gastrointestinal tract irritant and can cause nausea, vomiting, gastritis, and abdominal pain. Finally, isopropanol is directly toxic to myocytes and can induce severe hypotension, which is the strongest predictor of mortality in an isopropanol overdose.[90] Other systemic findings include hypoglycemia from impaired gluconeogenesis, hypothermia, and hemolytic anemia.

DIAGNOSTIC TESTING

Because specific measurement of toxic alcohols (by colorimetric and enzymatic assays or chromatographic methods) is not widely available, treatment should begin immediately if exposure to toxic alcohols is suspected.[105] EG and methanol poisoning should be considered in any patient presenting with a high AG metabolic acidosis and/or a high osmol gap, even in the absence of symptoms.[106] In early presentations, the osmol gap is more prominent, whereas in later presentations the high AG predominates. Because toxic alcohols are osmotically active compounds, the osmol gap, calculated as the difference between measured osmolality (by freezing point depression) and calculated osmolality, can be used as an approximation of the toxic alcohol concentration (in mmol/L), which can then be converted to mg/dL (Table 67.5). This estimation can be monitored serially during admission, and especially during dialysis, when precise serum concentrations are unavailable.[107,108] The calculated osmolality is based on the concentrations of sodium, glucose, blood urea nitrogen, and ethanol (if coingested), as follows:

Table 67.5 Conversion of Toxic Alcohol Concentration From mmol/L to mg/dL

Toxic Alcohol	Conversion From mmol/L to mg/dL
Acetone	×5.81
Diethylene glycol	×10.61
Ethanol	×4.61
Ethylene glycol	×6.21
Isopropanol	×6.01
Methanol	×3.20
Propylene glycol	×7.61

$$\begin{gathered}\text{Toxic alcohol concentration (in mmol/L)} \cong \text{osmol gap} = \\ \text{osmolality}_{\text{meas}} - \text{osmolality}_{\text{calc}} = \text{osmolality}_{\text{meas}} - \\ 2 \times \text{Na} + \text{gluc} + \text{urea} + 1.2 \times \text{ethanol; in SI units)} = \\ \text{osmolality}_{\text{meas}} - (2 \times \text{serum [Na]}) + (\text{glucose})/18 + \\ (\text{blood urea nitrogen})/2.8 + \text{ethanol}/4\text{; in Imperial units)}\end{gathered}$$

The normal osmol gap is between 0 and 10 mOsm/kg. An osmol gap higher than 10 suggests the presence of EG, methanol, isopropanol, propylene glycol, or acetone.[109] In isopropanol poisoning, both isopropanol and acetone contribute to the osmol gap.[110] However, it is important to recognize that a "normal" osmol gap does not exclude the diagnosis of toxic alcohol poisoning; if a specific patient has a baseline gap of 0, an excess of 25.6 mg/dL (8 mmol/L) of methanol would not result in an elevated osmol gap but would be clinically significant and warrant treatment. Also, the osmol gap would remain normal if a patient presents late after ingestion, once the parent alcohol has undergone complete oxidation, because glycolic acid and formic acid do not contribute to the osmol gap.[111]

Similarly, the AG can be used to estimate the glycolic acid and formic acid concentrations. The AG is the difference between measured cations (Na^+ and K^+) and measured anions (Cl^- and HCO_3^-), representing the difference between unmeasured anions and unmeasured cations (all values are in mmol/L). Lactate can also contribute to the AG and needs to be factored into the following equation:

$$\begin{gathered}\text{Glycolic acid/formic acid concentration} = \Delta\text{AG} - \Delta\text{lactate} = \\ ([\text{Na}+\text{K}] - [\text{Cl}+\text{HCO}_3] - 16\,[\text{AG upper reference limit}]) - \\ [\text{lactate}] - 2[\text{lactate upper reference limit}]) = \\ \text{Na}+\text{K}-\text{Cl}-\text{HCO}_3-\text{lactate}-14\end{gathered}$$

Studies have shown a good correlation between the concentration of metabolites and the calculated AG.[109] The absence of an AG suggests early presentation after EG or methanol exposure before metabolism into metabolites and coingestion of ethanol or another alcohol (e.g., isopropanol, propylene glycol).

Urinalysis can provide supporting evidence of EG exposure; calcium oxalate crystals (mono- and dihydrate forms) may be present in the urine sediment. These crystals appear 4 to 8 hours after ingestion and are found in approximately 50% of patients.[112,113] The presence of oxalate crystals is not specific for EG poisoning because it is also seen in individuals who ingest large amounts of vitamin C or a high-oxalate–containing food. Urine that fluoresces under Wood lamp illumination is another unique feature of EG poisoning. Many types of antifreeze contain sodium fluorescein, a fluorescent dye used as a marker to detect radiator leaks. Sodium fluorescein can be detected in the urine for up to 6 hours after ingestion.[114,115] Other laboratory abnormalities commonly found are hypocalcemia, leukocytosis, and an elevated protein concentration in cerebral spinal fluid.

Characteristic test findings of isopropanol exposure include increased osmol gap, the absence of metabolic acidosis (except if lactic acidosis is present), ketonemia, ketonuria, and normoglycemia. Acetonemia or acetonuria can be suspected by a positive sodium nitroprusside reaction in plasma or urine. A low concentration of serum ketones after 2 hours of isopropanol ingestion (in the absence of ALDH inhibition) generally excludes substantial ingestion.[92] Ketoacidosis from starvation, alcohol ingestion, or diabetes mellitus can be differentiated from isopropanol poisoning by the presence of metabolic acidosis.

TREATMENT

In case of toxic alcohol poisoning, rapid decision making is critical because most clinicians have to proceed without a toxic alcohol concentration confirmation. Management can be divided into supportive care, correction of the acidemia, antidote therapy, and enhanced elimination with ECTRs, although many of these are performed concomitantly.

Initial management, as for all poisonings, is directed toward stabilization and providing appropriate supportive care, which may include airway management, volume resuscitation, seizure management, and vasopressors. High minute ventilation should be maintained if significant acidemia is present. Because toxic alcohols are rapidly absorbed from the gastrointestinal tract, and because mucosal irritation is often present, gastrointestinal decontamination is seldom performed in this context. Aspiration of the gastric contents using a nasogastric tube may be useful if the ingestion is very recent.

The cornerstone of treatment for EG or methanol poisoning is to prevent their metabolism into toxic metabolites using an antidote (e.g., ethanol or fomepizole), which inhibits ALDH.

Indications for the use of these antidotes are as follows[106,116]:

1. Serum concentration of EG > 20 mg/dL (3.2 mmol/L) or methanol > 20 mg/dL (6.3 mmol/L)
2. Documented recent (hours) ingestion of toxic amount of EG or methanol and osmol gap > 10 mOsm/kg
3. A history or strong clinical suspicion of EG or methanol poisoning and at least two of the following: arterial pH < 7.3, serum bicarbonate < 20 mEq/L, osmol gap > 10 mOsm/kg, or presence of oxalate crystals in urine

Antidote treatment is not recommended for isopropanol poisoning because the metabolite acetone does not cause metabolic acidosis and is eliminated by endogenous routes. If fomepizole were used, the CNS depressant effect of isopropanol could even be prolonged.[117] Because ALDH has a greater affinity for ethanol than for methanol or EG, ethanol has been the traditional antidote used to prevent the formation of toxic metabolites and can be given enterally, intravenously, or via the dialysate. The intravenous formulation has the advantages of immediate bioavailability and avoiding gastrointestinal distress. Target serum ethanol concentrations for ADLH competition are between 100 and 150 mg/dL (21.7 and 32.5 mmol/L)[118]; below this concentration, ADLH inhibition would not maximally inhibit metabolite formation, whereas above this range, ethanol can exacerbate CNS and respiratory depression.[116] More predictable serum ethanol concentrations can be obtained during IHD by adding 475 mL of 65% ethanol to the 4.5-L acid bath of the dialysate.[118]

Fomepizole (4-methylpyrazole; Antizol) is a newer U.S. Food and Drug Association (FDA)–approved antidote for EG and methanol poisoning and has largely replaced ethanol in the United States. Compared with ethanol, fomepizole has multiple advantages: (1) more potent ALDH inhibition; (2) simple dosing; (3) predictable pharmacokinetics; (4) no blood monitoring; (5) few side effects and no CNS depression; and (6) longer duration of action. Its main drawback is the cost ($1000/dose in the United States). Either antidote is usually continued until EG or methanol concentrations are below 20 mg/dL (EG < 3.2 mmol/L or methanol < 6.3 mmol/L), and the patient is asymptomatic, with a normal arterial pH. Table 67.6 lists dosages of ethanol and fomepizole during toxic alcohol poisonings.

Acidemia should be corrected with the administration of IV sodium bicarbonate,[119] which enhances the deprotonation of acid metabolites, making them less likely to penetrate in end-organ tissues (e.g., retina, kidney) and more likely to be excreted by the kidneys. An initial intravenous bolus (1–2 mEq/kg), followed by an infusion, if necessary, should be given to maintain an arterial pH no less than 7.35. Asymptomatic hypocalcemia is not routinely treated in the setting of EG poisoning because it can potentially exacerbate calcium oxalate crystal formation and deposition.

Pyridoxine, thiamine, and magnesium are cofactors in the metabolism of EG, and their supplementation is recommended in patients who may be malnourished (e.g., alcoholics) or in those with known deficits.[106] Because the rate-limiting step of methanol metabolism is mediated by 10-formyltetrahydrofolate synthetase, which is folic acid—dependent, folic acid supplementation is recommended for the treatment of methanol poisoning. The suggested dose is 50 mg IV every 4 hours for 5 doses and then once daily.

Extracorporeal treatments, especially IHD, are extremely efficient at removing alcohols and their toxic metabolites, as well as rapidly correcting metabolic acidosis. When dialysis parameters are optimized (see earlier), the rate of clearance of alcohols and metabolites can reach 250 mL/min.

The increasing availability of fomepizole has modified the indications and pertinence of ECTR because of its great efficacy at preventing metabolite formation.[120] For example,

Table 67.6 Antidote Dosage During Toxic Alcohol Poisoning

Dose	Absolute Ethanol	10% IV Ethanol[b]	Fomepizole
Loading dose[a]	600 mg/kg	7.6 mL/kg	15 mg/kg IV
Maintenance dose	66 mg/kg per hour (nondrinker) 154 mg/kg per hour (chronic drinker)	0.8 mL/kg per hour (nondrinker) 2.0 mL/kg per hour (chronic drinker)	10 mg/kg q12h × 4 doses, then 15 mg/kg q12h
Maintenance dose during IHD	169 mg/kg per hour (nondrinker) 257 mg/kg per hour (chronic drinker)	2.1 mL/kg per hour (nondrinker) 3.3 mL/kg per hour (chronic drinker)	Same dose but q4h or a constant infusion of 1.0–1.5 mg/kg per hour

[a]Assumes initial ethanol concentration is zero; dose is independent of chronic drinking status.
[b]Equivalent to 7.9 g ethanol/dL.
IHD, Intermittent hemodialysis.

patients who are poisoned with EG but are neither acidotic nor have renal impairment may be treated with fomepizole alone, whatever the concentration of EG.[121] The same applies for methanol; however, the endogenous clearance of methanol is extremely low when fomepizole/ethanol is used. Assuming a methanol half-life of 54 hours under fomepizole,[122] a patient with an initial methanol concentration of 320 mg/dL (100 mmol/L) would need to be hospitalized 9 days until the methanol is at a safe concentration (<20 mg/dL, or 6.3 mmol/L). Dialysis might therefore be instituted to reduce hospitalization costs and antidote requirement for patients poisoned with EG or methanol.[67,123] However, IHD is mandatory when there is metabolic acidosis or an increased AG because these are suggestive of toxic metabolites accumulation.

Indications for ECTR for EG and methanol poisoning include the following[15]:

- Serum EG or methanol concentration > 50 mg/dL (8.0 mmol/L, or 15.6 mmol/L, respectively) if fomepizole is not used
- Metabolic acidosis (pH < 7.2) or an AG > 28 mmol/L
- Coma or seizures
- Vision deficits secondary to methanol
- AKI or chronic kidney disease (CKD)

The expected duration of dialysis in hours can be estimated using these formulas and using SI units:

$$4.7 \times (\text{Ln}\,[\text{initial EG concentration}/2])$$ [124]

or

$$3.5 \times (\text{Ln}\,[\text{initial methanol concentration}/4])$$ [125]

provided that the metabolic acidosis is corrected. However, serial monitoring of the toxic alcohol concentration or osmol gap is recommended to confirm the estimation.[126] IHD should be continued until the parent alcohol levels are < 20 mg/dL (EG < 3.2 mmol/L or methanol < 6.3 mmol/L), and the metabolic acidosis is corrected. Other modalities, such as CRRT, would offer lower clearance and require longer treatment but may be considered if IHD is unavailable.[127]

IHD can also effectively remove isopropanol and acetone, although it is only indicated when the isopropanol concentration is > 400 mg/dL (66.6 mmol/) or there is prolonged coma, hypotension, myocardial depression, tachydysrhythmias, or AKI.[128–130] Other modalities would not offer comparable clearances.[131]

Phosphate addition in the dialysate is often required during IHD. The use of heparin should be minimized or preferably avoided in methanol-poisoned patients due to the increased risk of intracerebral hemorrhage.

SALICYLIC ACID

Salicylates are widely used as analgesics and antiinflammatory medications. Salicylic acid produces its antiinflammatory effects by suppressing the expression of cyclooxygenase (COX), thereby reducing the production of proinflammatory mediators such as prostaglandins. In overdose, salicylic acid also uncouples oxidative phosphorylation, which is a key contributor to toxicity and death. Aspirin (acetylsalicylic acid) is commonly prescribed as an antiplatelet therapy. Salicylic acid is used as a topical keratolytic agent and wart remover, bismuth subsalicylate (Pepto-Bismol; 236 mg of salicylate/15 mL) is used for reflux disease, and methyl salicylate (oil of wintergreen; 98% salicylate; 1 teaspoon contains 7 g of salicylates) is used for pain relief and as a flavoring agent.[132,133] Because there are many over-the-counter formulations, salicylates are among the substances most commonly involved in exposure calls to poison centers.[1] Fortunately, because of modification in packaging and the availability of nonsteroidal antiinflammatory drug (NSAID) alternatives, significant salicylate poisonings have become less frequent in the United States.

TOXICOLOGY AND TOXICOKINETICS

Salicylates, in general, are rapidly absorbed by the gastrointestinal tract. Peak serum concentrations are reached within 1 hour, unless enteric-coated products are used. However, in an acute overdose, bezoar formation and pylorospasm may delay absorption and the appearance of symptoms.[134,135]

Within the therapeutic range, salicylic acid (molecular weight = 138.1 Da; V_D = 0.2 L/kg) is 90% protein-bound and undergoes first-order hepatic metabolism. Normally, less than 10% of salicylate is excreted unchanged by the kidneys, and its elimination half-life is between 2 and 4 hours.[136,137] Salicylic acid is filtered at the glomerulus, actively secreted in the proximal tubule, and reabsorbed passively in the distal tubules. In case of an acute overdose, the protein binding falls to 50%, the V_D increases, and major pathways of metabolism become saturated, so clearance changes from first-order to zero-order. Although more salicylic acid is eliminated by the kidneys in these circumstances, the other dose-dependent changes result in a marked increase in the elimination half-life (>30 hours).[138–140]

Salicylic acid is a weak acid, with a pK_a of 3.0. It exists in an ionized and a nonionized state in the plasma:

$$H^+ + Sal^- \leftrightarrow HSal$$

Uncharged particles, such as unionized salicylic acid, cross the blood-brain barrier and other tissues more readily than the charged form. Acidemia drives the above equilibrium to the right, therefore causing more CNS toxicity.[141] Elimination is similarly influenced by the urinary pH, where an increase in tubular lumen pH drives the equilibrium toward the ionized form, limits its uptake by renal tubular cells, and favors its elimination in urine. This provides the rationale for urinary alkalinization to enhance the elimination of salicylates.

CLINICAL PRESENTATION AND DIAGNOSTIC TESTING

Salicylate poisoning can be acute or chronic. Acute ingestions >150 mg/kg usually present with mild to moderate toxicity; >300 mg/kg, patients are at risk of severe poisoning, and exposures >500 mg/kg are potentially lethal. The therapeutic range is 10 to 30 mg/dL (0.7–2.2 mmol/L); concentrations > 40 mg/dL (2.9 mmol/L) can be associated with toxicity, and concentrations > 75 mg/dL (5.4 mmol/L) are associated with significant severe effects.

Acute salicylate ingestion often causes nausea and vomiting as a result of gastritis and direct stimulation of the chemoreceptor trigger zone in the medulla. Hemorrhagic ulcers, decreased gastric motility, and pylorospasm are also seen. A variety of acid–base abnormalities may occur with salicylate poisoning, but the classic finding is mixed respiratory alkalosis and high AG metabolic acidosis. Salicylate stimulates the respiratory center in the brainstem independently of the aortic and carotid chemoreceptors, leading to an early fall in CO_2 partial pressure and respiratory alkalosis.[142–144] Subsequently, metabolic acidosis is induced by the accumulation of salicylate and organic acids. Increased minute ventilation promotes lactic acid production.[142,145] In addition, salicylates uncouple mitochondrial oxidative phosphorylation and interrupt glucose and fatty acid metabolism in the Krebs cycle, leading to an increase in the production of tissue carbon dioxide, lactic acid, and ketoacids.[145]

Salicylates cause a variety of CNS effects, either directly or through the selective reduction of the brain glucose concentration. Cerebral edema, perhaps secondary to capillary leak, may also play a role in alterations in mental status.[145] Neurologic manifestations include tinnitus, central hyperthermia, vertigo, altered mental status (e.g., hyperactivity, agitation, delirium, hallucination), and coma.[146,147] As stupor progresses, there may be blunting of the respiratory response, which may further decrease the pH and increase salicylate entry into the CNS.[141,148] Tinnitus occurs at salicylate concentration of 30 mg/dL (2.2 mmol/L) and may lead to decreased auditory acuity and even deafness.[149] Early salicylate poisoning may present with hyperglycemia resulting from glycogenolysis, gluconeogenesis, and decreased peripheral use. Hypoglycemia occurs later, with heightened cellular energy demand and uncoupling of oxidative phosphorylation.[150]

Chronic poisoning may be seen in patients who receive prolonged salicylate therapy, often in the setting of reduced renal clearance. Symptoms in this setting are often more prominent than after an acute ingestion at the same salicylate concentration; such patients are often misdiagnosed as having delirium, encephalopathy, or fever of unknown origin, and they have a high mortality.[151,152] Noncardiogenic pulmonary edema, a classic albeit rare finding, may occur as a result of gastrointestinal local release of vasoactive peptides and increased capillary permeability, which further limits the application of urine alkalinization.[153,154]

The diagnosis of salicylate intoxication is usually suspected from the history, classic clinical findings, and metabolic abnormalities (described earlier). An elevated AG with concomitant respiratory alkalosis should prompt confirmation of salicylate exposure. Quantitative serum salicylate concentrations can generally be determined rapidly in many centers. Because absorption may be erratic or prolonged, serial measurements (every 2–4 hours) are initially required. The magnitude of the concentration is less important in patients with significant symptoms because treatment will be initiated regardless. In these cases, the salicylate concentration is most useful for monitoring the effectiveness and determining the duration of therapy. The Done nomogram, which was an attempt to correlate salicylate concentrations with toxicity, is no longer used in practice because of its poor predictive value.[155]

TREATMENT

General principles of poisoning management apply to salicylates. Special consideration should be geared toward respiratory support. Patients are dependent on maintaining a high minute ventilation and high serum pH to reduce salicylate entry into the CNS. Endotracheal intubation should therefore only be performed if absolutely necessary, and by an experienced clinician, to avoid prolonged periods of apnea, during which many deaths have been reported.[156] Ventilator settings should try to replicate the patient's respiratory pattern prior to intubation, although this is usually difficult because of auto–positive end-expiratory pressure (PEEP).

Once the patient is stabilized, further therapy is aimed toward decreasing absorption and increasing elimination of salicylates. Activated charcoal remains the preferred decontamination technique.[157,158] MDAC can enhance the elimination of salicylates but is considered less efficacious and more cumbersome to perform compared with urine alkalinization.[5,6,159,160] Nevertheless, MDAC should be considered, especially for patients who may have a significant amount of unabsorbed salicylate in the gastrointestinal tract.

Serum and urine alkalinization is a crucial component of treatment. As mentioned previously, alkalinization will drive salicylate to be dissociated, which will reduce its diffusion through the blood-brain barrier and its tubular reabsorption (ion trapping). Because the ionization constant (pK_a) is a logarithmic function, small changes in urine will have a large effect on salicylate elimination.[161] Urinary clearance of salicylate is enhanced several-fold with alkalinization when compared with forced diuresis alone. In one small series, the percentage of the salicylate dose excreted in urine increased from 2% under acidic conditions to 30% under alkaline conditions.[162] The bicarbonate infusion is titrated to reach a urinary pH of 7.5 or until the salicylate concentration is below 30 mg/dL (2.2 mmol/L). An alkaline urine cannot be produced in the presence of severe hypokalemia because kidney reabsorption of potassium occurs via the H^+/K^+ exchange pump in the collecting tubule, so the potassium level should be monitored and aggressively corrected.

Alkalinization may be contraindicated in patients with AKI or pulmonary edema; in these cases, IHD may be preferred over the risk of precipitating respiratory failure and requiring mechanical ventilation. Acetazolamide induces urinary alkalinization but is absolutely contraindicated because it lowers the arterial pH, which promotes salicylate movement into the CNS and other tissues.

The first article ever published on diffusion-based techniques—in 1913, by John Abel—showcased the removal of salicylates from animal subjects.[163] Salicylate displays properties of a highly dialyzable compound because of its low V_D, small molecular size, and low protein binding at high concentrations.[164–166] IHD is the best extracorporeal treatment because it can remove salicylate while correcting the acid–base and volume status of salicylate-poisoned patients. Although peritoneal dialysis,[167,168] exchange transfusion,[155] HP,[169–171] and CRRT[172–174] all increase salicylate clearance, none of them achieve the efficacy of IHD. IHD remains underused in salicylate poisoning; the availability of alternative treatments (e.g., MDAC, urinary alkalinization) may lure the clinician into a false sense of security, so most deaths still occur before ECTR is initiated.[76]

Indications for ECTR include any of the following attributable to salicylate poisoning[16]:

- Neurologic symptoms (e.g., confusion, seizures, coma)
- Pulmonary edema
- pH < 7.25
- Serum salicylate concentration > 90 mg/dL (6.5 mmol/L)
- Acute kidney injury
- Clinical deterioration, despite appropriate treatment

Patients with chronic poisoning display more toxicity at lower salicylate concentrations. Therefore, in chronic poisoning, the initiation of ECTR is based largely on signs and symptoms. ECTR should be maintained until the salicylate concentration is below 19 mg/dL (1.4 mmol/L) and there has been a clinical improvement. Although some authors have suggested continuing urinary alkalinization during ECTR, dialysis would likely alkalinize a patient much more quickly and reliably than an intravenous infusion of bicarbonate.[175]

LITHIUM

Lithium, the lightest metal on the periodic table, was used therapeutically for gout in the 19th century and later in soft drink preparations. Its use became widespread in the 1950s, when it became a first-line therapy for the treatment of bipolar disorder.[176] Although generally considered a safe drug, lithium has a narrow therapeutic range (0.7–1.0 mmol/L) and can induce major side effects when serum concentrations become supratherapeutic. Lithium may also cause long-term metabolic and renal effects, such as dysthyroidism, hyperparathyroidism, nephrogenic diabetes insipidus, and a progressive decline in the glomerular filtration rate (GFR), which are beyond the scope of this chapter.[177] Its mechanism of action and toxic effects are incompletely understood. Lithium is thought to stabilize cell membranes, reduce neural excitation, and reduce synaptic transmission. Potential mechanisms include depletion of CNS inositol,[178] inhibition of the intracellular signaling pathways involved in neuroprotection,[179,180] and modulation of nitric oxide, glutamate, and other neurotransmitters.[181]

TOXICOLOGY AND TOXICOKINETICS

The pharmacokinetic parameters of lithium are well known. Lithium is a 7 Da monovalent cation, orally administered as a carbonate (capsule) or citrate (liquid). Oral absorption is rapid and complete at therapeutic doses, and its bioavailability is not affected by food. Peak blood concentrations with therapeutic dosing are reached within 1 to 2 hours for an immediate-release formulation and within 4 to 6 hours for the sustained-release form, but they are usually delayed several-fold in acute poisoning. Lithium is unbound to proteins and has a V_D of 0.7 to 0.8 L/kg. Its distribution into various tissues is variable—lithium diffuses into the liver and kidneys rapidly, but its transfer into bone, muscle, and brain is much slower, which explains the delay in peak CNS concentrations relative to the serum concentration after an acute overdose.[182] Lithium also predominantly distributes into the intracellular compartments by active transport. It is eliminated almost exclusively by the kidney, where it is freely filtered; 80% of filtered lithium is reabsorbed (75% in the proximal tubule and 25% in the distal tubule). Total body clearance is therefore approximately 20% of the GFR. Lithium reabsorption follows that of sodium, and therefore sodium-avid states (e.g., volume depletion, NSAIDs, congestive heart failure, cirrhosis) markedly increase lithium retention. The half-life of lithium with therapeutic use is about 18 to 24 hours for normal subjects but can be prolonged in older adults, chronic lithium users, and CKD.[183,184] However, in patients with normal renal function, in an acute overdose the plasma elimination half-life is approximately 10 hours (largely due to distribution into extravascular tissues), whereas in a chronic overdose it is 30 to 40 hours.[185]

CLINICAL PRESENTATION AND DIAGNOSTIC TESTING

Lithium overdose may be defined as acute when it occurs after a single massive exposure in a patient naïve to lithium and as chronic when it occurs after dosing or prescribing errors or in situations where lithium clearance becomes impaired.[186] The severity of symptoms does not correlate closely to the serum lithium concentration. Acutely poisoned patients may be completely asymptomatic at a lithium concentration of 4.0 mmol/L, whereas there may be evident clinical signs in chronic poisonings with concentrations near the therapeutic range.[184,187] Acute poisoning is predominantly manifested by gastrointestinal symptoms (e.g., nausea, vomiting, diarrhea) and by nonspecific cardiac conduction delay, although life-threatening dysrhythmias are uncommon. Neurologic findings are especially prominent in chronic poisoning and may range from mild symptoms (e.g., coarse tremor or dysarthria to a more severe presentation, such as lethargy, seizures, hyperthermia, coma, and death).[188,189] A protracted neurologic course is sometimes seen after severe poisoning,[183] and some patients develop a syndrome of irreversible lithium-effectuated neurotoxicity (SILENT), which relates to cerebellar and cognitive deficits that may last years. Causative factors and their relation to lithium is unknown.[190]

TREATMENT

Therapy should not only be guided by lithium concentrations but also by symptoms, underlying kidney function, and a patient-specific risk assessment.[185,187] Initial supportive care should target the specific manifestations of lithium toxicity, including treatment of hyperthermia, dysrhythmias, and seizures. Volume contraction favors proximal lithium reabsorption and should therefore be promptly corrected. Although isotonic saline (0.9%) is preferred for volume resuscitation, it may need to be subsequently replaced by hypotonic solutions or free water if lithium-induced nephrogenic diabetic insipidus and hypernatremia become a concern.

Gastrointestinal decontamination may be required in massive oral ingestions, although oral activated charcoal does not bind lithium and has no role in isolated lithium poisoning.[191] Whole-bowel irrigation with polyethylene glycol can be considered for sustained release formulations.[192] Oral sodium polystyrene sulfonate (Kayexalate), a cation exchanger commonly used for the treatment of hyperkalemia, has been shown to bind unabsorbed lithium from the gastrointestinal tract and enhance elimination of absorbed lithium in animals and humans.[23,193,194] It should be considered in patients who have mild to moderate symptoms for whom dialysis is delayed or not considered as a treatment option.[23]

Lithium has ideal properties for extracorporeal removal—small size, negligible protein binding, small V_D, low endogenous clearance. However, it is unknown if enhancement of lithium removal by ECTR translates into clinical benefit; in one retrospective and underpowered comparative study, clinical outcomes were similar between groups treated or untreated with IHD, although cohorts were not comparable at baseline.[195]

IHD is the modality of choice when extracorporeal elimination is required. Clearances in excess of 180 mL/min can be obtained with modern filters.[196–199] Serum lithium concentrations often rebound after IHD termination,[183,200] but as mentioned earlier, this may not be concerning, because lithium CNS concentrations actually decrease during redistribution back to the circulation[80] unless there is ongoing absorption of lithium from the gut. CRRT provides inferior clearance and removal rates than IHD.[197,201–203] Lithium clearance with PD is inferior to that of functioning kidneys.[204,205]

Indications for ECTR include any of the following[12]:

- Severe neurologic features (central hyperthermia, seizures, and/or depressed consciousness)
- Serum [Li] > 5 mmol/L (although good outcomes have been noted from much higher concentrations in acute poisoning without ECTR)
- Kidney impairment with symptoms and serum [Li] > 4 mmol/L
- Life-threatening cardiac dysrhythmias

The threshold for dialysis initiation should be lower in patients who cannot tolerate volume repletion or those with impaired kidney function.[183,206]

VALPROIC ACID

Valproic acid (VPA) is used for the treatment of absence seizures, complex partial seizures, migraine, and mood disorders. Although acute VPA intoxication often results in mild self-limiting CNS depression, serious toxic effects and death have been reported.[207,208]

TOXICOLOGY AND TOXICOKINETICS

VPA (molecular weight, 144.21 Da; V_D, 0.2 L/kg) is available in immediate and sustained-release formulations, both of which have a high bioavailability. Serum concentrations typically peak 1 to 13 hours after ingestion, depending on the preparation.[209] Therapeutic serum concentrations range from 50 to 100 mg/L (347–693 μmol/L). Protein binding of VPA is saturable in overdose. Protein binding is typically 90% at therapeutic concentrations, but decreases to 35% at 300 mg/L (2079 μmol/L).[210]

VPA is rapidly metabolized by the liver. It undergoes glucuronic acid conjugation (70%) and beta and omega oxidation to various metabolites, whereas less than 3% is normally excreted unchanged in urine.[210] In overdose, more of its metabolism undergoes CYP450-mediated omega oxidation, the metabolites of which are thought to be responsible for some of the toxic effects of VPA, such as 5-OH-VPA and 4-en-VPA.

CLINICAL PRESENTATION AND DIAGNOSTIC TESTING

Most VPA overdoses are well tolerated, but toxicity is more likely following ingestions over 200 mg/kg.[208,211] Acute poisoning is typically manifested by gastrointestinal distress (e.g., nausea, vomiting, diarrhea), CNS abnormalities (e.g., confusion, obtundation, coma with respiratory failure), hypotension, and elevated transaminase concentrations. Free (unbound) and total VPA serum concentrations are poorly correlated with severity of intoxication, but most patients with total concentrations greater than 180 mg/L (1247 μmol/L) develop some degree of CNS depression.[212]

Hyperammonemia is seen at therapeutic and toxic valproate concentrations[213] and does not usually cause toxicity but, when markedly elevated, it can cause encephalopathy, cerebral edema, and death. The mechanism of valproate-induced hyperammonemia is incompletely understood, but may relate to the inhibition of carbamoyl phosphate synthetase, *N*-acetylglutamate synthetase, or carnitine-dependent beta oxidation, which inhibits the urea cycle.[213]

At very high serum concentrations (>1000 mg/L [6930 μmol/L]), complications include high AG metabolic acidosis, elevated osmolality (VPA concentration > 1500 mg/L [10,395 μmol/L] may raise the osmol gap by 10 mOsm/kg or more), hypernatremia, hypocalcemia, pancreatitis, noncardiogenic pulmonary edema, bone marrow suppression, and AKI.[214,215] Diagnosis of VPA intoxication is based on a history of exposure, typical toxic symptoms, and an elevated serum valproate concentration.

TREATMENT

Treatment consists of initial stabilization of respiratory and cardiovascular function. Gastrointestinal decontamination using activated charcoal should be administered, particularly if the patient presents within 1 hour of exposure. Due to the prolonged absorption noted in overdose, it may also be useful beyond that period. Urine elimination enhancement is ineffective because the excretion of VPA by the kidney is limited. L-Carnitine has been proposed as a treatment for hyperammonemic encephalopathy but supporting data are limited.[216] If used, intravenous L-carnitine should be administered with a loading dose of 100 mg/kg, followed by infusions of 50 mg/kg every 8 hours, until ammonia levels are decreasing.

The low molecular weight and V_D of VPA are favorable for extracorporeal elimination. Although IHD has little effect on the elimination of VPA at usual serum concentrations because of its extensive protein binding, significant clearance can be obtained at supratherapeutic drug concentrations when binding to plasma proteins is saturated.[209,217] IHD has the added advantage of clearing ammonia and reversing metabolic acidosis.[209]

Indications for ECTR include any of the following attributable to valproate toxicity[14]:

- Serum valproate concentration > 900 mg/L (6250 μmol/L)
- Cerebral edema or shock
- Coma or respiratory depression requiring mechanical ventilation
- Acute hyperammonemia
- pH ≤ 7.10

Rebound of VPA concentrations is often observed 5 to 13 hours following cessation of high-flux IHD, requiring additional sessions.[210] Charcoal HP has been successfully used in some cases but is limited by early column saturation.[207] Tandem, or IHD-HP in series, may be the most effective technique but probably offers only a marginal advantage over IHD to offset the added cost and potential complications of the technique.[218] Intermittent HDF has been used successfully in two reports,[219,220] but there is insufficient evidence to determine whether this technique offers additional benefit over IHD alone. CRRT appears considerably less effective than intermittent alternatives and should only be used if IHD is unavailable.[14,221] Albumin dialysis, slow low-efficiency dialysis with filtration, TPE, and PD are inferior therapeutic options in valproate poisoning and have not been recommended.[222,223]

CARBAMAZEPINE

Carbamazepine is a widely used anticonvulsant agent that is also being used increasingly for pain management and mood disorders.

TOXICOLOGY AND TOXICOKINETICS

Carbamazepine (molecular weight, 236 Da) and tricyclic antidepressants have a similar chemical structure. Its therapeutic effect results from binding to sodium channels, inhibiting neuronal depolarization, and decreasing glutamate release. It also has anticholinergic effects at high concentrations.

Carbamazepine is available in immediate- and modified-release formulations and is characterized by erratic and incomplete absorption, which is exacerbated in overdose due to pharmacobezoar formation and ileus secondary to its anticholinergic properties.[224] It is lipophilic and has a V_D of 1.2 L/kg and a protein binding of approximately 75%, which does not decrease much in overdose.[225] It undergoes hepatic metabolism, mainly through CYP450 3A4 into many metabolites, the most important being carbamazepine-10,11-epoxide, which is pharmacologically active. Carbamazepine induces its own metabolism, which increases endogenous clearance with chronic use. The therapeutic concentration range of carbamazepine is 4 to 12 mg/L (16.9–50.8 μmol/L).

CLINICAL PRESENTATION AND DIAGNOSTIC TESTING

Carbamazepine toxicity frequently presents with neurologic, cardiovascular, and anticholinergic symptoms, which may be delayed in onset because of its erratic absorption. Mild toxicity (≅ 30 mg/L [127 μmol/L]) presents as drowsiness, nystagmus, tachycardia, hyperreflexia, or dysmetria. In more severe exposures (>40 mg/L [169 μmol/L]), lethargy, seizure, coma, QRS prolongation, hypotension, and pronounced anticholinergic symptoms (especially ileus) may develop; death is unusual. Agranulocytosis and syndrome of inappropriate antidiuretic hormone (SIADH) are associated with chronic use and are not typically seen in acute poisonings.

The diagnosis of carbamazepine toxicity relies on the history and presence of typical clinical findings and is confirmed by laboratory testing. Serial measurements of serum carbamazepine concentrations are required because the time to peak concentration may be significantly delayed, including beyond 24 hours. Concentrations should be obtained every 4 to 6 hours until a definite downward trend is seen.

TREATMENT

Most patients poisoned with carbamazepine can be managed with supportive care alone, including fluids, ventilatory support, benzodiazepines for seizure control, vasopressors for hypotension, and sodium bicarbonate for sodium channel blockade. Gastrointestinal decontamination is administered, even if the patient presents late after ingestion due to the prolonged absorption phase, but is contraindicated if ileus is present. No antidote exists to reverse carbamazepine's toxicity. MDAC can enhance carbamazepine clearance and may even reduce the duration of coma and need for mechanical ventilation[5,226] but, again, is contraindicated in the presence of ileus.[5,226–229]

Severe carbamazepine poisonings can be treated using ECTR, which can provide better and more predictable clearance than MDAC, particularly when there is an ileus.[230–232] Historically, charcoal or resin HP was the ECTR of choice because of carbamazepine's extensive protein binding. However, with high-flux filters, high blood flows, and larger catheters, carbamazepine clearance with HP and IHD are comparable, exceeding 100 mL/min.[231–239] IHD is preferred due to its greater availability, lower cost, and lower complication rate compared with HP. There are limited data on the ECTR clearance of the toxic metabolite carbamazepine-10-11-epoxide, but it also appears to be dialyzable—its protein binding is less than that of carbamazepine (50% vs. 75%–90%).[225,236,240,241] CRRT, TPE, and albumin dialysis have been used in carbamazepine poisoning but do not provide comparable removal rates.[242,243]

Indications for ECTR include any of the following attributable to carbamazepine toxicity[11]:

- Prolonged coma
- Seizures, cardiovascular instability, symptoms unresponsive to supportive care
- Rising concentrations despite MDAC
- Serum carbamazepine concentration > 45 mg/L (190 μmol/L)

BARBITURATES

Barbiturates are CNS-depressants used as sedatives, hypnotics, anxiolytics, and anticonvulsants. Barbiturates were extremely popular until the availability of benzodiazepines in the 1960s. Although their use has steadily decreased over the years, barbiturates, especially those that are long-acting agents, are still a concern.[1] Phenobarbital is the most commonly available barbiturate worldwide and is most often implicated in poisoning, although others are seen in developing countries.[244]

TOXICOLOGY AND TOXICOKINETICS

Most barbiturates are weak acids, with a pK_a of approximately 7. Their two principal mechanisms of action are potentiating the effect of the gamma-aminobutyric acid (GABA) receptor and blocking the α-amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid (AMPA) receptor (a subtype of glutamate receptor); both contribute to the CNS effects.[245] They are usually categorized by their duration of action: short (>3–4 hours; pentobarbital, secobarbital), intermediate (>4–6 hours; amobarbital, butabarbital), or long-acting (≥6–12 hours; barbital, primidone, phenobarbital).[244] Absorption is variable and is influenced by the dose ingested, presence of an ileus, and concomitant ingestion of other drugs. The ability of a particular barbiturate to penetrate the blood-brain barrier reflects its lipophilicity and determines its clinical effects. Barbiturates are small molecules with a protein binding usually below 50%, although long-acting barbiturates are less protein-bound than short-acting ones. The metabolism of barbiturates is mostly hepatic, but there may be renal excretion for those that are less lipophilic, such as phenobarbital. There may be enzymatic induction and higher barbiturate clearance for patients on chronic therapy.

The following discussion focuses on phenobarbital because it is the most commonly encountered, although it shares properties similar to other long-acting and short-acting barbiturates. Recommendations also apply to primidone because it is metabolized to phenobarbital.[246] An oral dose of 1 g of most barbiturates will cause serious toxicity in most adults, whereas death can result with ingestions of more than 2 g[247] or with a concentration higher than 80 mg/L (345 μmol/L)[248] in the absence of supportive care.

CLINICAL PRESENTATION AND DIAGNOSTIC TESTING

Phenobarbital has clinical effects that are prolonged.[249,250] In mild exposures, toxicity usually manifests as an altered level of consciousness. Moderate poisonings will produce apnea and hypotension. More severe cases present with areflexia, cutaneous bullae, circulatory collapse (although more common with short-acting barbiturates), hypothermia, and coma.[244,247] Concomitant AKI, cardiac, or pulmonary disease may increase the clinical sensitivity to barbiturates.[251] Early deaths after barbiturate ingestion are caused by respiratory and cardiovascular collapse, whereas delayed deaths are caused by acute lung injury, ventilator-acquired pneumonia, cerebral edema, or multiorgan system failure. Specific serum testing assays of barbiturates other than phenobarbital are not widely available.

TREATMENT

Supportive measures, including rewarming, hydration, and vasopressors, are usually sufficient in most cases of barbiturate poisoning. Patients may be profoundly comatose and require prolonged mechanical ventilation. There is no antidote for barbiturates.

If an airway is secured, MDAC can facilitate the clearance of phenobarbital.[252] Because barbiturates are weak acids, urinary alkalinization can also increase phenobarbital renal clearance by at least two- to threefold.[253,254] However, because renal clearance of phenobarbital is already low (<3 mL/min), alkalinization would have little effect on total body clearance.[253] MDAC has a greater impact on clearance than alkalinization.[254,255] Surprisingly, the duration of coma was shown to be reduced by urine alkalinization in one study[256] but not by MDAC in a small randomized trial despite a decreased half-life.[257] Another controlled study has suggested that MDAC alone is superior to either urinary alkalinization alone or MDAC plus urinary alkalinization.[255] Therefore, MDAC is preferred to urine alkalinization in the absence of ileus.[5,6,244]

An animal study has shown a very significant and clear decrease in mortality in dogs and rats that underwent HP after a lethal phenobarbital infusion[164] compared with those who did not. No randomized studies have evaluated the effect of ECTR in humans, although the mortality rate appeared lower in an uncontrolled group that underwent HP.[258] HP and IHD are particularly appealing treatments for toxicity from long-acting barbiturates because the endogenous clearance is low, especially in patients who are naïve to barbiturates.[244] The endogenous elimination half-life for barbiturates other than phenobarbital, barbital, and primidone is short and therefore less likely to benefit from extracorporeal removal. Modern high-flux dialyzers provide clearances at least equal to those of HP.

Indications for ECTR include any of the following (in particular for phenobarbital poisoning)[244]:

- Coma with respiratory depression
- Hypotension not responding to vasopressors
- Inefficiency of MDAC at reducing serum concentrations

A caution with the treatment of barbiturate poisoning is the risk of precipitating barbiturate withdrawal in chronic users. This occurs when concentrations fall below the therapeutic range and can manifest as seizures and/or persistent delirium after approximately 48 to 72 hours of treatment.[70,259–261] This can be minimized by stopping ECTR and other treatments when the phenobarbital concentration is therapeutic; ongoing monitoring can promptly detect rebound toxicity or withdrawal, especially for short-acting barbiturates and chronic users. In the case of phenobarbital, it may also be prudent to reinitiate barbiturate treatment when the serum concentration is therapeutic and clinical recovery has been observed.

PHENYTOIN

Phenytoin is used as a first-line treatment of epilepsy, both for status epilepticus and seizure prevention. Phenytoin toxicity is relatively frequent considering its narrow therapeutic range, although phenytoin-related deaths are rare.

TOXICOLOGY AND TOXICOKINETICS

Phenytoin stabilizes neuronal membranes and decreases seizure activity, possibly by producing a voltage-dependent blockade of membrane sodium channels implicated in the action potential leading to seizures.[262] Phenytoin has a molecular weight of 252 Da and high protein binding (90%–95%), which decreases to 70% in the presence of renal failure or hypoalbuminemia, although surprisingly remains almost unchanged in overdose. Only the unbound fraction of phenytoin has a biologic effect.

Phenytoin is available in various dosing forms and has erratic oral bioavailability,[263] especially in overdose.[264] Phenytoin is metabolized predominantly by hepatic hydroxylation through CYP2C9 and is then eliminated in the bile as an inactive metabolite, reabsorbed from the intestinal tract and excreted in urine. Elimination is first-order at low serum concentrations but becomes zero-order (saturable) at higher concentrations, but within the reference range. The elimination half-life of oral preparations ranges from 14 to 22 hours. Therapeutic serum concentrations range between 10 and 20 mg/L (40–80 μmol/L). The lethal dose in adults is estimated at 2 to 5 g.

CLINICAL PRESENTATION AND DIAGNOSTIC TESTING

Most of the toxic manifestations of phenytoin are neurologic; their severity is loosely correlated to its serum concentration; between 20 and 40 mg/L (80–160 μmol/L), symptoms include nystagmus, ataxia, dysarthria, and mild CNS depression.[265] Over 40 mg/L (160 μmol/L), lethargy, confusion, hypotension, coma, and seizures may be observed. Death is usually caused by respiratory or circulatory depression but is rare. Cardiovascular toxicity is unusual with oral formulations, but short-lived atrioventricular delays and bradycardia may be occasionally seen with rapid intravenous infusion.[265]

Total serum phenytoin concentrations should be determined in all suspected overdose cases. Some conditions may alter the protein-binding capacity, such as uremia, extremes of age, and concomitant use of agents that displace phenytoin from its albumin binding site (e.g., salicylates, sulfonamides, tolbutamide, VPA). The free phenytoin concentration can be determined in such cases and, in the context of hypoalbuminemia, can assist with interpretation; toxicity is apparent when its free level is more than 2.1 mg/L (8.3 μmol/L).[266]

TREATMENT

Treatment of phenytoin poisoning is primarily supportive because most patients have an excellent outcome, although the duration of admission can be prolonged. Benzodiazepines can be given for seizures. Gastrointestinal decontamination can reduce the severity of toxicity and limit the systemic burden of phenytoin, MDAC significantly reduces the elimination half-life of phenytoin in healthy and poisoned patients[22,267,268] and is commonly used, particularly in patients in whom serum phenytoin concentrations are increasing or remain elevated.[5] There is no specific antidote for phenytoin poisoning.

Experience with ECTRs remains anecdotal and limited. In theory, given the high protein binding of phenytoin, HP or TPE has been historically favored, and they both has been used successfully in this context.[269–275] Surprisingly, despite phenytoin's high protein binding, there is some evidence that IHD may also enhance its elimination, perhaps due to its low binding constant to albumin, which ensures a constant pool of freely diffusible unbound phenytoin.[276–278] This finding is attributable to newer high-flux, high-efficiency dialyzers—older obsolete dialysis apparatus did not have any effect on phenytoin clearance.[279]

The combination of HP and IHD could potentially maximize clearance and has been used in a few studies,[274,280,281] although it remains unclear if the combination is superior to either technique used alone. Neither peritoneal dialysis nor CRRT has any role in phenytoin poisoning,[282–285] whereas the data are still uncertain for albumin dialysis.[59] In summary, the outcome of most phenytoin poisonings is favorable, although the duration of admission can be prolonged by many days. Supportive care alone is usually sufficient. In the case of actual or anticipated (e.g., persistently elevated phenytoin concentration) prolonged neurologic toxicity, IHD or HP can be considered.[18]

METFORMIN

Metformin is a first-line drug for the treatment of non–insulin-dependent diabetes, particularly in overweight patients. Metformin improves insulin sensitivity in patients and is now the most popular antidiabetic drug in the world. The related biguanide, phenformin, was withdrawn in 1978 because of the high incidence of severe lactic acidosis.

TOXICOLOGY AND TOXICOKINETICS

Metformin is a small unbound molecule (165 Da) that has a large volume of distribution (3 L/kg) due to its ability to diffuse into the intracellular compartment and bind to microsomes. Its bioavailability is incomplete. It does not undergo hepatic metabolism and is eliminated unchanged by renal tubular secretion (renal clearance ≅ 500 mL/min with normal renal function); hence, caution is required when prescribing it to patients with kidney impairment. The toxic dose in acute ingestions is not well established but is likely to exceed 100 mg/kg.

CLINICAL PRESENTATION AND DIAGNOSTIC TESTING

Metformin toxicity can manifest in various ways—gastrointestinal symptoms (e.g., abdominal pain, diarrhea, nausea, vomiting), lactic acidosis, and hypotension are all hallmarks of metformin intoxication. Hypoglycemia, hypothermia, altered mental status, and acute pancreatitis have also been reported.[286,287] Metformin-associated lactic acidosis (MALA) usually occurs in the presence of underlying conditions, particularly AKI or CKD. MALA is defined as an arterial pH ≤ 7.35 and lactate concentration > 45 mg/dL (5 mmol/L)[288,289]; it can result from acute or chronic poisoning. There is controversy in regard to the association between metformin and lactic acidosis; some sources have claimed that the association is coincidental and due to other factors, such as sepsis and heart failure,[290,291] whereas others have suggested that metformin itself directly causes lactic acidosis.[292] The finding of asymptomatic patients who develop toxicity shortly after an acute toxic ingestion of metformin appears to give credence to the latter hypothesis[293–295] and is occasionally named "metformin-induced lactic acidosis" (MILA).[292] The true prevalence of MALA and MILA is unknown, but severe life-threatening acidosis is estimated at a rate of 0.05 cases/1000 patient-years.[296]

The physiopathology of MALA is not entirely understood but seems to be related to interference with hepatic gluconeogenesis and impaired lactate utilization. Metformin also binds to mitochondrial membranes, which in overdose can shift energy utilization toward anaerobic metabolism, leading to the production of lactate.[297] The increase in lactic acid production is then compounded by the defect in lactate clearance by the kidneys or liver (mentioned earlier).

Because serum metformin measurements are not usually available in a clinically useful time frame, the diagnosis must be suspected in every patient taking metformin who presents with severe lactic acidosis. Some studies have reported a correlation between metformin concentrations and markers of lactic acidosis,[290,292,298] but not all.[299] A range of factors may influence the relationship between metformin concentrations and clinical outcomes, which have been discussed elsewhere.[292,300]

TREATMENT

Given the absence of a specific antidote to metformin, the management of metformin toxicity is supportive care—normalizing the acid–base status, reducing the metformin concentration, and treating concomitant and exacerbating conditions. Decontamination with activated charcoal should be considered after an acute large ingestion of metformin in patients presenting within 1 to 2 hours of ingestion. Patients who develop lactic acidosis, hypoglycemia, or other signs of metformin toxicity should be admitted to an intensive care unit to facilitate treatment.

Severely acidotic patients (pH ≤ 7.1) should receive intravenous sodium bicarbonate, although treatment can be limited by hypernatremia and volume overload. ECTRs have a definite role in the treatment of MALA because they can correct acidosis faster than intravenous bicarbonate, contribute to the clearance of lactate and metformin, and treat volume overload and uremia. A dramatic improvement from ECTR has been described in some acutely poisoned patients soon after dialysis initiation, suggesting that the benefit might be more attributable to pH correction than metformin removal.[301] One study compared patients who underwent dialysis with those who did not, and the mortality rate did not differ between the two groups. However, the IHD group was sicker at baseline.[288] There is some evidence that ECTRs can remove a significant amount of metformin, especially when kidney function is impaired,[302–304] and also because metformin distributes to a smaller apparent V_D in AKI; thus it is more available for extracorporeal removal. In terms of case reports of ECTR in the treatment of poisoning, metformin is the leading intoxication reported in the recent literature.[305]

Indications for ECTR include any of the following attributable to metformin toxicity[19]:

- Lactate concentration > 180 mg/dL (20 mmol/L)
- Arterial pH ≤ 7.1
- Failure of supportive therapy in severe MALA
- The presence of shock, impaired kidney function, or coma

IHD is preferred over CRRT due to higher metformin and lactate clearances[19,74,306]—for example, lactate removal is greater following 6 hours of IHD compared with 24 hours of CVVHDF,[307] although it is likely that lactate removal by ECTR is much less than that by physiologic processes that are functioning normally.[308] Endpoints for ECTR cessation include normalization of blood pH and lactate level. Shortened sessions can result in a marked rebound in lactic acidosis,[309] which may possibly be reduced by using IHD initially, followed by CRRT.[310] HP is not recommended for treatment of MALA without IHD because it will not correct acid-base abnormalities.[310]

PARAQUAT

Paraquat (1,1′-dimethyl-4-4′-bipyridylium dichloride) is a cheap, nonselective, fast-acting herbicide that is inactivated on contact with soil, which reduces environmental contamination. Because it is not readily absorbed across intact skin, and because the droplets generated from the aerosol spray are large, toxicity from direct contact or inhalation is rare. However, accidental or intentional oral ingestion of paraquat is extremely toxic, with high rates of morbidity and mortality (50%–90%). Its use and access are restricted in many parts of the world due to its high toxicity, but it is available in many developing countries.[311]

TOXICOLOGY AND TOXICOKINETICS

Paraquat (molecular weight, 186 Da; protein binding, 5%) has an oral bioavailability that is less than 30%. Peak concentrations are generally reached 2 hours after ingestion, but maximal tissue distribution occurs during the next 6 hours.[312,313] Paraquat distributes to most organs, especially the lungs, kidneys, and liver, with a $V_D \cong 1.0$ L/kg. It is predominantly excreted by the kidneys, so that the elimination half-life is 12 hours with normal renal function, but it can increase to more than 48 hours with impaired renal function.[313,314]

Paraquat is directly toxic to mucosal membranes, causing corrosion and esophageal perforation. Paraquat catalyzes the formation of reactive oxygen species (ROS), in particular superoxide radicals, through redox cycling. Because this process can be sustained by the extensive supply of electrons and oxygen in the lungs, the resultant oxidative stress causes extensive cell injury and necrosis and a secondary inflammatory reaction.[312,315,316]

CLINICAL PRESENTATION AND DIAGNOSTIC TESTING

The presentation and clinical course of paraquat poisoning depend largely on the dose ingested and timing of presentation to a health care facility after exposure. Ingestions < 20 mg/kg may not cause symptoms other than oral ulceration, vomiting, and diarrhea. Ingestions between 20 and 40 mg/kg will usually cause multiple organ failure, with death occurring between 1 and several weeks postexposure. Patients who ingest over 40 mg/kg usually die within 3 days from accelerated multiorgan failure and/or from the corrosive effects of paraquat.[317–320]

Most paraquat-related deaths are caused by respiratory failure. Types I and II alveolar epithelial cells take up paraquat via an energy-dependent polyamine transporter.[312,321] Patients develop an acute respiratory distress syndrome (ARDS)–like presentation, which ultimately progresses to irreversible pulmonary fibrosis. In the kidneys, paraquat causes acute tubular necrosis, which may be severe, but will resolve if the patient survives the acute phase of poisoning.[322,323] Other clinical manifestations include severe upper gastrointestinal tract ulcerations, hepatic injury, and shock. Paraquat does not readily cross the blood-brain barrier, and therefore does not typically cause CNS effects; seizures are rarely reported.

Serum paraquat concentration predicts the mortality risk, and nomograms have been validated to correlate with clinical outcome.[324–326] Unfortunately, few laboratories have the ability to perform a quantitative paraquat assay.[326–329] A rapid and inexpensive qualitative test for urine paraquat can be performed by adding sodium dithionite to urine. If the color of urine changes from yellow to blue, it confirms the presence of recent paraquat exposure (the more intense the blue, the higher the paraquat exposure).

TREATMENT

Any suspected paraquat exposure warrants prompt assessment and aggressive management. After initial stabilization, volume resuscitation should be provided. Oxygen should be used only if required for hypoxemic patients because it may promote cellular toxicity from paraquat but, in such patients, the prognosis is likely to be very poor. Paraquat is absorbed quickly, which limits the efficacy of the usual decontamination techniques, although activated charcoal should be administered for presentations within 1 hour of ingestion; paraquat-induced caustic injury is not usually noted for many hours postingestion. Many paraquat formulations already contain a proemetic, and some contain an alginate to reduce absorption.[330] Forced diuresis does not increase the renal clearance of paraquat,[314] and no specific antidote is available.

Because paraquat is a small molecule, unbound to protein, and with a reasonably small volume of distribution, its content in plasma is removable by standard ECTRs (clearance > 120 mL/min). However, paraquat distributes quickly to tissues, especially to the lungs, where it is trapped and accumulates. Furthermore, once the toxic process of free-radical generation is established, it is unlikely that any elimination enhancement could lessen its progression. This concept has been confirmed in animal studies; 50% of the dogs exposed to a lethal dose of paraquat survived with HP when it was initiated 2 hours after exposure, whereas 100% mortality occurred when it was initiated at 12 hours.[331]

IHD or HP is mostly reserved for patients presenting early after ingestion, especially within 4 hours.[332] In that scenario, all resources should be pooled to initiate ECTR as quickly as possible.[333] Other ECTRs, including peritoneal dialysis, are of no use.[314] Rebound is usually observed following ECTR treatment.[314,334–339]

The poor prognosis related to paraquat poisoning has prompted the use of several experimental treatments, largely based on immunosuppression, or other types of antiinflammatories. Unfortunately, none has been proven to alter clinical outcomes significantly. Examples include vitamins C and E, superoxide dismutase, deferoxamine, selenium, niacin, *N*-acetylcysteine, thiosulfate, salicylate, corticosteroids, cyclophosphamide, and radiotherapy.

Of note, the combination of cyclophosphamide and corticosteroids failed to identify a statistically significant benefit,[340] although high-dose corticosteroids remain recommended. The palliative care team should also be integrated into the treatment plan early, given the disastrous outcome of most patients.

THEOPHYLLINE

Theophylline is a methylxanthine bronchodilator used in the treatment of asthma, chronic obstructive pulmonary disease, and infant apnea. Its use has largely declined over the years in favor of inhaled corticosteroids, anticholinergics, and β_2-adrenergic agonists.[341]

TOXICOLOGY AND TOXICOKINETICS

Theophylline (molecular weight, 180.2 Da; protein binding, 40%) acts as a nonselective inhibitor of phosphodiesterase and a nonselective antagonist of the adenosine receptor. This results in smooth muscle relaxation, catecholamine release, bronchodilation, peripheral vasodilatation, and inotropic and chronotropic activation.[342]

Theophylline is available in immediate- and slow-release formulations and is completely absorbed by the gastrointestinal tract. It distributes in a volume smaller than that of water (0.5 L/kg). Theophylline is predominantly metabolized in the liver by CYP1A2, and only 10% is excreted by the kidneys. Endogenous metabolism is generally slow, but the pharmacokinetics and metabolism of theophylline are influenced by age, gender, body weight, concurrent illness (e.g., congestive heart failure, liver disease, infection), cigarette smoking (increases clearance), and medications known to induce or inhibit the CYP1A2 isozyme.[343] At the therapeutic range (10–20 mg/L [55–110 μmol/L]), theophylline exhibits first-order kinetics; however, at toxic concentrations, the kinetics change to zero order, and a small increase in dosage can lead to a dramatic increase in serum concentration.[344]

CLINICAL PRESENTATION AND DIAGNOSTIC TESTING

Theophylline intoxication may result from acute ingestion (e.g., deliberate self-poisoning or medication error) or chronic use (e.g., prescribing or dispensing error; decrease in clearance from a drug interaction, smoking cessation, or intercurrent illness). The distinction is important because for the same serum theophylline concentration, symptoms are more severe from chronic toxicity than those following acute intoxication.[345] The therapeutic index is very narrow; even at therapeutic concentrations, nearly one-third of patients may exhibit signs of mild intoxication. Patients are likely to demonstrate toxicity at serum concentrations > 30 mg/L (167 μmol/L), and severe toxicity is likely when it is >90 mg/L (500 μmol/L). Caffeine, which is an active metabolite of theophylline, shares many of its toxic effects.

Early symptoms of toxicity consist of transient caffeine-like effects, such as nausea, vomiting, diarrhea, irritability, tremor, headache, insomnia, and tachycardia. Patients with moderate intoxication may be lethargic and disoriented and may develop supraventricular tachycardia and frequent premature ventricular contractions. Severe intoxication is potentially life-threatening and includes symptoms such as seizures, hyperthermia, hypotension, ventricular tachycardia, rhabdomyolysis, and AKI. Irreversible brain injury may follow seizures if they are not prevented or treated rapidly. Death most often results from cardiorespiratory collapse or hypoxic encephalopathy following cardiac dysrhythmias or generalized seizures.[345–347] Hypokalemia, caused by β_2-adrenergic–induced transcellular potassium shifts, can be severe and is most often seen following acute overdose.[346,348]

TREATMENT

Aggressive supportive measures are required for severe theophylline poisoning. Hypotension usually responds to volume repletion and vasopressors. Tachydysrhythmias can be corrected by β-adrenergic antagonists, such as propranolol or esmolol, but should be used with caution in patients susceptible to bronchospasm.[349,350] Theophylline-associated seizures can be particularly difficult to control; benzodiazepines are first-line therapy, but propofol, barbiturates, and even neuromuscular paralysis may be required in refractory cases, at which point the prognosis is poor. MDAC increases the clearance of theophylline but is sometimes limited by profound and intractable emesis in severely intoxicated patients.[351–353] Intubation and ventilation can be performed for the purpose of preventing vomiting to enable MDAC treatment.

Because of its low molecular weight, low protein binding, low endogenous clearance, and low V_D, theophylline is an ideal candidate for extracorporeal elimination. Extracorporeal clearances can surpass 150 mL/min.[354–356] There are numerous reports of successful treatment with HP and IHD. In one observational retrospective cohort study, a group treated with HP or IHD had a significantly shorter duration of clinical toxicity compared with a group managed with supportive care only, despite being sicker at presentation.[357] IHD was shown to provide similar clearance to HP, with fewer complications in one retrospective study[42] and, because it also corrects hypokalemia, IHD is the preferred treatment.

Continuous techniques have also been reported in the management of theophylline intoxication with acceptable results, despite their inferior clearances.[358,359] Peritoneal dialysis and TPE are not useful alternatives. Exchange transfusion is an option for neonates.[360]

Indications for ECTR include any of the following attributable to theophylline toxicity[13]:

- Clinical deterioration despite optimal therapy
- Serum theophylline concentration > 100 mg/L (555 µmol/L)
- Chronic poisoning with serum theophylline concentration > 60 mg/L (333 µmol/L)
- The presence of refractory seizures, shock, life-threatening dysrhythmias, inability to administer charcoal because of intractable vomiting, or extremes of age (<6 months or >60 years)
- A rising serum theophylline concentration despite optimal therapy, or if MDAC cannot be administered

The ECTR should be continued until clinical improvement is achieved, and the serum concentration is less than 15 µg/mL (83 µmol/L). Rebound is usually minor but should be monitored after ECTR in case there is also ongoing absorption.

ACETAMINOPHEN

Acetaminophen (*N*-acetyl-*p*-aminophenol, also known as paracetamol) is an analgesic and antipyretic available without prescription in various forms and combinations. Acetaminophen is a major cause of deliberate self-poisoning.

TOXICOLOGY AND TOXICOKINETICS

The actions of acetaminophen remain poorly defined, but may include the inhibition of COX in the brain. Analgesic and antipyretic activity occurs at serum acetaminophen concentrations between 10 and 20 mg/L (66–132 µmol/L).[361] Acetaminophen is available in immediate- and extended-release preparations and has high oral bioavailability. Serum concentrations usually peak within 4 hours of ingestion, but may be delayed by many hours with massive overdoses or from sustained-release formulations.

Protein binding (≅20%) and volume of distribution (0.8 L/kg) of acetaminophen are small and not known to change in overdose. Acetaminophen undergoes extensive hepatic metabolism, in which 90% is conjugated to inactive metabolites and the remainder is oxidized, mostly by CYP2E1 to *N*-acetyl-*p*-benzoquinoneimine (NAPQI). Glutathione combines with NAPQI to form a nontoxic complex that is eliminated in the urine. In overdose, however, there is an accumulation in NAPQI and glutathione depletion, allowing NAPQI to bind covalently to proteins in hepatocytes, inducing injury and liver cell necrosis.[362] The lowest toxic dose following acute poisoning is 10 g in a single dosing over 8 hours.[363]

CLINICAL PRESENTATION AND DIAGNOSTIC TESTING

Acetaminophen poisoning cases are difficult to recognize at the early stages. Patients may experience nonspecific symptoms, including nausea, vomiting, anorexia, malaise, or abdominal pain, in the first 24 hours. Hepatic injury, according to an increase in transaminase levels, typically occurs 12 to 24 hours after ingestion, by which time the efficacy of *N*-acetylcysteine is reduced. The severity of hepatotoxicity varies with acetaminophen dose, nutrition, and other medical conditions and is usually detected by an elevation in serum transaminase levels (aspartate aminotransferase [AST] and alanine transaminase [ALT]), which peak 2 to 3 days after ingestion in mild to moderate poisoning but later in severe hepatotoxicity. Hepatotoxicity is defined as a peak AST above 1000 IU/L.[364] More severe cases can evolve to fulminant liver failure with encephalopathy, coagulopathy, hypoglycemia, and death for those who cannot undergo liver transplantation. AKI may also occur because of tubular susceptibility to NAPQI by various possible mechanisms, which include the CYP450 pathway, as well as prostaglandin synthetase and *N*-deacetylase enzymes.[365] AKI can persist beyond the duration of hepatotoxicity but generally recovers after 1 to 3 months.

The assessment of the risk associated with acute acetaminophen poisoning can be largely estimated by timing the exposure to a serum acetaminophen concentration using the Rumack-Matthew nomogram (or a locally applicable version).[366] A point below the treatment line indicates that the patient has a low risk of developing hepatotoxicity. This nomogram has been validated, despite some limitations (which are outside the scope of this chapter). For example, a serum acetaminophen concentration of 150 mg/L (1000 µmol/L) at 4 hours postingestion indicates the need for *N*-acetylcysteine (NAC) treatment.

TREATMENT

A comprehensive management of acetaminophen toxicity is reviewed elsewhere.[367] Most cases are triaged and treated in the emergency department in collaboration with the regional poison control center and intensivists, without the involvement of a nephrologist. Management of acetaminophen toxicity is mainly focused on decontamination for early presentations and timely administration of the antidote, NAC.[368] NAC serves as a glutathione donor, increasing the capacity to detoxify NAPQI.[369] It is most efficacious when initiated within 8 hours of an acute overdose but is also useful if initiated later. NAC is prescribed if there is evidence of hepatic injury following a history of known or possible acetaminophen poisoning or if the acetaminophen concentration is above the treatment line on the Rumack-Matthew nomogram.

Although acetaminophen is amenable to extracorporeal removal (low V_D, small molecule, low protein binding), ECTR is rarely necessary, given the safety, low cost, efficacy, and availability of NAC. IHD or CRRT can be considered for patients with severe AKI according to the usual criteria or occasionally in those presenting following massive ingestions with a pattern of mitochondrial toxicity—coma, lactic metabolic acidosis, and/or cardiovascular instability.[8,370,371] These patients can be identified by early onset of these signs in the context of a massive exposure, as opposed to the features of hepatotoxicity due to the delayed presentation, as described earlier. In these cases, the potential risk of ECTR removing NAC and compromising its effect can be overcome by at least doubling its rate of infusion (some sources have suggested that an increased dose of NAC is required with massive acetaminophen poisoning anyway).[69,372] Liver replacement therapies such as the molecular adsorbents recirculating system (MARS) and single-pass albumin dialysis (SPAD) have been used for liver support as bridges to liver transplantation or pending native liver recovery.[373–377] However, limited availability of these techniques, as well as major bleeding complications associated with them and their high cost,warrant further investigation.

METHOTREXATE

Methotrexate is a folate analog used to treat a variety of cancers and rheumatologic and dermatologic diseases. Methotrexate inhibits dihydrofolate reductase, an enzyme necessary for the synthesis and replication of RNA and DNA.

TOXICOLOGY AND TOXICOKINETICS

The oral absorption of methotrexate is limited by a saturable intestinal absorption mechanism for doses over 30 mg/m^2. Recent research has suggested that a single acute oral ingestion of methotrexate is unlikely to have any toxic potential in patients with normal renal function.[378]

Methotrexate toxicity is most often related to dosing errors with parenteral administration or repeated supratherapeutic chronic dosing. Methotrexate (molecular weight, 454 Da) has a small V_D (0.4–0.8 L/kg) and is 50% bound to plasma proteins, regardless of serum concentration. Methotrexate is primarily excreted unchanged (80%–90%) in the urine by passive glomerular filtration and active tubular secretion; its renal clearance varies greatly and decreases at higher doses. It also undergoes hepatic and intracellular metabolism into active metabolites.

In the presence of renal failure, methotrexate can rapidly accumulate in the serum and tissue cells. A toxic concentration of methotrexate is defined as greater than 5 to 10 μmol/L at 24 hours, 1 μmol/L at 48 hours, and 0.1 μmol/L at 72 hours.[379,380] These concentrations are predictive of renal, gastrointestinal, mucosal, and bone marrow toxicity in patients receiving chemotherapy. Methotrexate concentrations in patients being treated for indications other than cancer should not surpass 0.01 μmol/L.[381]

CLINICAL PRESENTATION AND DIAGNOSTIC TESTING

At high concentrations, methotrexate and its metabolites can precipitate in renal tubules, causing crystal nephropathy and AKI. Methotrexate can also cause bone marrow suppression, mucositis and stomatitis, liver damage, and neurotoxicity. The onset of the toxicity is generally rapid; nausea and vomiting typically begin 2 to 4 hours after high-dose therapy (>1000 mg/m^2). CNS toxicity usually follows approximately 12 hours after high-dose intravenous methotrexate therapy or intrathecal administration. Mucositis and pancytopenia will manifest approximately 1 to 2 weeks after exposure.

TREATMENT

Activated charcoal may be given early after presentation after an acute oral overdose, although toxicity in this scenario is unusual if kidney function is normal.[378] Aggressive hydration is indicated for patients presenting with methotrexate toxicity. Because methotrexate is a weak acid ($pK_a = 5$), and because it is eliminated largely unchanged in the urine, urinary alkalinization with intravenous sodium bicarbonate (target urine pH ≅ 8) will enhance methotrexate elimination and may help prevent AKI.[6,382]

Leucovorin (folinic acid, an active form of folate) can limit the bone marrow and gastrointestinal toxicity of methotrexate by bypassing the effects of methotrexate on dihydrofolate reductase. Leucovorin rescue is most beneficial when administered promptly after methotrexate exposure and should always be given to patients after a supratherapeutic exposure. Other supportive measures include transfusion of blood components for established cytopenias, antiemetics, and nutritional support for stomatitis. Dose recommendations and precise application are covered elsewhere.[383]

Glucarpidase (carboxypeptidase G_2; $CPDG_2$) is a recombinant bacterial enzyme that catabolizes methotrexate to inactive metabolites. It is used in combination with leucovorin because glucarpidase cannot access intracellular stores of methotrexate. It decreases the serum methotrexate concentration within 1 hour following administration. This FDA-approved antidote is used in patients with a serum methotrexate concentration > 1 μmol/L with prolonged methotrexate clearance (AKI) or under investigational protocol for intrathecal (IT) overdoses (100 mg IT methotrexate). Because it is expensive (>$50,000) and is unavailable or restricted in most countries, it may not be readily available for each case of methotrexate toxicity.

IT overdoses of methotrexate may require special measures, including cerebrospinal fluid drainage and exchange and administration of corticosteroids, in addition to leucovorin-glucarpidase. Methotrexate can be removed by extracorporeal treatments; high clearances can be achieved by various modalities, especially charcoal HP and high-flux IHD.[379] CRRT, as expected, provides lesser removal rates,[384] whereas PD and TPE are ineffective.[385–387] ECTR clearance is usually below endogenous clearance in patients with intact kidney function and is therefore most beneficial for patients with AKI. The indication for ECTR, especially with the arrival of glucarpidase, remains to be defined but may be related to cost and availability. ECTR seems to be indicated if serum methotrexate concentrations are above the following: 1600 to 2200 μmol/L at the end of infusion, 30 to 300 μmol/L at 24 hours, 3 to

30 μmol/L at 48 hours, and over 0.3 μmol/L at 72 hours. ECTR protocols are usually continued until methotrexate concentrations are less than 0.1 μmol/L.[388] Rebound in serum methotrexate concentrations, attributable to redistribution, often follows ECTR.

OTHER FACTORS

There are other conditions and poisoning situations for which enhanced elimination using ECTR may reduce the duration of toxicity. Although we cannot present them in detail here, there have been reports of successful extracorporeal removal of the following poisons: aluminum,[389] *Amanita phalloides*,[390] baclofen,[391] barium,[392] bromate,[393] carambola,[394] cefepime,[395] cibenzoline,[396] dabigatran,[78,397,398] dapsone,[399] diethylene glycol,[400] fluoride,[401] gabapentin,[402] gentamicin,[403] isoniazid,[404] metronidazole,[405] pregabalin,[406] propylene glycol,[407] thallium,[7] and vancomycin.[408]

CONCLUSION

General supportive care is sufficient to manage most poisoned patients. In a small selection of cases, extracorporeal blood purification, usually consisting of IHD, can help reduce the exposure of a patient to the toxic effects of a poison. An understanding of poison toxicokinetics can help a clinician discern the timely conditions and circumstances when ECTRs are most likely to be beneficial.

Complete reference list available at ExpertConsult.com.

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BOARD REVIEW QUESTIONS

1. What is the extracorporeal treatment most used in the context of poisonings?
   a. Intermittent hemodialysis
   b. Intermittent hemoperfusion
   c. Continuous renal replacement therapy (CRRT)
   d. Plasmapheresis

   Answer: a

   Rationale: Hemodialysis provides the best clearance for all poisons aside from those that are extensively protein-bound. Hemodialysis is widely available, has a lower cost than the other modalities, and has minimal complications.

2. Which of the following poisons is not amenable to extracorporeal removal?
   a. Carbamazepine
   b. Lithium
   c. Amitriptyline
   d. Phenobarbital

   Answer: c

   Rationale: Extracorporeal removal is not effective due to the large volume of distribution of amitriptyliine.

3. Why has the use of hemoperfusion in poisonings decreased over the years?
   a. It is more expensive than hemodialysis.
   b. Cartridges are not as available.
   c. It has a greater complication rate than hemodialysis.
   d. The advent of new dialyzers has rendered dialysis at least as efficient as hemoperfusion for the majority of poisonings.
   e. All of the above.

   Answer: e

   Rationale: Hemoperfusion cartridges also need to be changed every 2–3 hours because of saturation. Finally, hemoperfusion cannot correct electrolyte disorders and does not bind metals or alcohols.